

			DAILY DRILLING REPORT				REPORT # 001		21/08/2018	
							SPUD: 21/08/2018		DAY # 047	
WELL	Palm Valley 13	24:00 MD/ TVD	23.5 / 23m BRT	DRILLED	16m	AFE DAYS / COST	45	\$11,347,521		
RIG	ENSIGN 932	FORMATION	Hermannsburg Sandstone	DAILY COST	\$129,022	CUM.	\$5,259,127			
DAILY OP'S SUMMARY		Completed rig up and prespud inspections, drilled and installed mousehole, spudded PV13 drilled to 23.5mMD and ran conductor #1.								
FORECAST OP'S		Waited on Haliburton cementers to reset hours, cement conductor, wait on cement, install riser, make up BHA and Drill 17 1/2" hole.								
LAST CASING	508mm	20.0"	SET AT	23.5m	MAASP		BOP TEST	NIL	DUE	
LAST L.T.I. 430 Days		SAFETY					WEATHER: Day		Fine	
		1. PTSM; Daily work scope, BBS cards, PTW, JSA and prespud check lists.					Night		Fine	
		2. PTSM; Discussed making up BHA, housekeeping, grader working on road and prespud documents.								

BIT INFORMATION				BHA # 1		MUD PROPERTIES			OPERATION		HRS	CUM
WOB(Klb)	3-10	JET V(fps)	98	TOOL	LENGTH	Mud Type	Gel Spud Mud		1 Move/ Rig Up/Down	15.50	1119.50	
RPM	60	H S I	0.03	Bit - Tri Cone	0.53	Depth (m)	8	18	2 Drilling	2.00	2.00	
BIT NUMBER	1	A		Bit Sub	0.76	Density (ppg)	8.45	9.05	3 Wash / Ream			
Size (inches)	24.0	24.0		NOV Fluid Hammer- 9 5/8"	10.99	ECD (ppg)	8.54	9.14	4 Coring			
Make	Stealth	CONDUCTOR		Crossover Sub	0.76	Temp (° C)		20	5 Circ & Condition	2.00	2.00	
Type	STC42X			Drill Collar- 9 1/2"	9.29	Viscosity (sec)	106	74	6 Tripping- Bit / Casing	0.75	0.75	
IADC Code	425X			Crossover Sub	1.09	PV / YP (cp/lb)	22 / 34	14 / 29	6.1 Tripping- Other			
Serial Number	HT9011			Crossover Sub	0.66	Gels (10s/m)	28 / 34	21 / 31	6.2 Wiper Trip			
T.F.A. (in)	1.052			Crossover Sub	0.32	API Filt. (cc)	15.0	11.5	6.3 Handle BHA			
Depth In (m)	8					Sand (% Vol)	0.0	2.0	7 Service / Slip Line			
Depth Out (m)	24	8				KCl (%)			8 Repairs			
Total Meters	16	8				pH (strip)	9.5	10.0	10 Survey			
Hours IADC OnBtm	2 2	0 0				LGS %/Vol			11 Logging			
ROP IADC OnBtm	8 8					Chlorides (ppm)	160	260	12 RU for Csg / Cmt			
Condition Out	1 1 WT A E 0 No TD					Surface Vol. (Bbls)	373		12.1 Run Casing	2.75	2.75	
FLOW DATA				BHA LENGTH	24.40	Hole Vol.	20		12.2 Cementing	1.00	1.00	
RATE (gpm/lps)	321	20.2		BHA WEIGHT	14.92	Mud Mixed	436		13 Wait on Cement			
AV - DP (fpm/mpm)	14	4		STRING WT	14.88	New Hole Drilled	28.5		14 BOP's / Wellhead			
AV - DC (fpm/mpm)	16	5		HOOK LOAD	42.00	Downhole Losses	63		15 Test BOP / Formation			
SPP (psi/atm)	630	43		WT BELOW JARS		Surface Losses	-28		19 Tight hole / Fishing			
SPP (Calculated)	630	43		HOURS: JAR MTR	2.0	CHEMICAL USAGE			21 Completion			
PUMP DATA				DRAG: UP DOWN	0 0	AMC Biocide G	2		21.1 Well Control			
#1: NAT 8-P-80	STROKE	9.0"		TORQUE: ON OFF	1500 3600	AMC Bore Seal Fine	10		21.2 Other			
RATE				SURVEYS: MD INC' AZ'		AMC Quickseal Medium	30		TOTALS	24.00	8.50	
LINER	6.0"			0 0 0		Aus-Ben A	140		NON PRODUCTIVE	1.00	433.00	
#2: NAT 8-P-80	STROKE	9.0"				Caustic Soda	3		PRODUCTS: USED REC ON SITE			
RATE	100					Kwickseal Fine	14		DIESEL (L)	2150	47750	
LINER	6.0"					Xan-Bore	1		BARITE (T)			
#3: IDECO T-800	STROKE	8.5"							DRILL WATER	8000	58000	
RATE									DAILY MUD COSTS \$4,792.24			
LINER	6.0"								CUM. MUD COSTS \$4,792.24			

HOURLY OPERATIONS SUMMARY 0000 to 2400		
From	To	[IADC Code] Description
0:00	6:00	[1] Mixed spud mud and LCM pill, surface lines pressure tested to 250/2500 psi - OK, made up BHA and prepared to drill mousehole.
6:00	9:00	[1] Drilled mouse hole and installed mouse hole sock.
9:00	10:00	[1] Broke down and laid out 12 1/4" BHA.
10:00	11:00	[1] Reset rig smart and crown-o-matic set points.
11:00	12:00	[1] PJSM, made up 24" BHA.
12:00	12:30	[1] Held PJSM, discussed handling BHA, Driving to the conditions with Grader on road.
12:30	14:30	[1] Made up 24" BHA, Centralize BHA in sub-base for hole centre.
14:30	15:30	[1] Conducted Muster drill and 1 man missing drill, Conducted Pre-spud hazard hunt, Held pre-spud safety meeting with all personnel.
15:30	17:30	[2] Spudded PV13 and drilled 24" hole from 8m to 23.5m RT
17:30	17:45	[5] Circulated hole clean.
17:45	18:00	[6] Pulled out of hole.
18:00	19:45	[5] PJSM, pumped out and cleaned out cuttings from cellar.
19:45	20:15	[6] PJSM, racked back BHA and stood in mast.
20:15	23:00	[12.1] PJSM, picked up and ran 20" conductor casing, made up and ran cement stringer.
23:00	0:00	[12.2] NPT- Waited on Haliburton cementers to reset hours.

HOURLY OPERATIONS SUMMARY 0000 to 0600 on 22/08/18		
From	To	Description
0:00	3:00	Waited on Haliburton cementers to reset hours.
3:00	4:30	Rigged up to cement, PJSM, mixed and pumped 3.5 bbls 15.8 ppg cement plug through stinger inside casing.
4:30	6:00	Waited on cement. Laid out stinger.

POB-36: Ensign - 20, CT - 4, OICS - 3, Halliburton - 4, Pason - 1, OCS - 1, SiteMed - 1, RMN - 1, ERMS - 1								9,852 Hrs		
RECEIVED: Drill water and mist pumps.						DISPATCHED:				
MAXIMUM GAS:			BACKGROUND GAS:			CONNECTION GAS:			TRIP GAS:	
SUPERVISOR: Kev Dau / Mark Booth			GEOLOGIST: Phil Allen / Peter Morris			RIG MGR: Gary Hender / Adam delaine				

			DAILY DRILLING REPORT			REPORT # 002		22/08/2018	
						SPUD: 21/08/2018		DAY # 048	
WELL	Palm Valley 13	24:00 MD/ TVD	23.5 / 23m BRT	DRILLED		AFE DAYS / COST	45	\$11,347,521	
RIG	ENSIGN 932	FORMATION	Hermannsburg Sandstone		DAILY COST	\$111,556	CUM.	\$5,370,682	
DAILY OP'S SUMMARY		Waited on Halliburton cementers to reset hours, cemented conductor, waited on cement, install riser, make up 17 1/2" BHA and repaired mud pump. One incident reported - pinched finger injury.							
FORECAST OP'S		Complete mud pump repairs and drill 17 1/2" hole.							
LAST CASING	508mm	20.0"	SET AT	23.5m	MAASP		BOP TEST	NIL	DUE
LAST L.T.I. 431 Days		SAFETY 1. Discussed Safety cards, cementing operations, hazard hunt, daily work scope. 2. Discussed PTW system, Hazard identification, working at heights identifying correct fall restraints.					WEATHER: Day Fine Night Fine		

BIT INFORMATION			TOOL		LENGTH		MUD PROPERTIES		OPERATION		HRS	CUM		
WOB(Klb)		JET V(fps)					Mud Type	Gel Spud mud	1 Move/ Rig Up/Down			1119.50		
RPM		H S I					Depth (m)	24	2 Drilling			2.00		
BIT NUMBER	1	A					Density (ppg)	8.70	3 Wash / Ream					
Size (inches)	24.0	24.0					ECD (ppg)		4 Coring					
Make	Stealth	CONDUCTOR					Temp (° C)		5 Circ & Condition			2.00		
Type	STC42X						Viscosity (sec)	65	6 Tripping- Bit / Casing			0.75		
IADC Code	425X						PV / YP (cp/lb)	12 / 30	6.1 Tripping- Other					
Serial Number	HT9011						Gels (10s/m)	14 / 16	6.2 Wiper Trip					
T.F.A. (in)	1.052						API Filt. (cc)	13.0	6.3 Handle BHA	2.00		2.00		
Depth In (m)	8						Sand (% Vol)	0.1	7 Service / Slip Line	0.50		0.50		
Depth Out (m)	23.5	8					KCl (%)		8 Repairs	3.50		3.50		
Total Meters	15.5	8					pH (strip)	10.0	10 Survey					
Hours IADC OnBtm	2 2	0 0					LGS %/Vol		11 Logging					
ROP IADC OnBtm	7.8 7.8						Chlorides (ppm)	260	12 RU for Csg / Cmt					
Condition Out	1 1 WT A E 0 No TD						Surface Vol. (Bbls)	1159	12.1 Run Casing			2.75		
FLOW DATA			BHA LENGTH			Hole Vol.			26	12.2 Cementing	5.50	6.50		
RATE (gpm/lps)						Mud Mixed			794	13 Wait on Cement	5.00	5.00		
AV - DP (fpm/MPM)						New Hole Drilled				14 BOP's / Wellhead				
AV - DC (fpm/MPM)						Downhole Losses			8	15 Test BOP / Formation				
SPP (psi/atm)						Surface Losses				19 Tight hole / Fishing				
SPP (Calculated)						CHEMICAL USAGE				21 Completion				
PUMP DATA			DRAG: UP DOWN			AMC Biocide G			2	21.1 Well Control				
#1: NAT 8-P-80			STROKE			9.0"			Aus-Ben A	49	21.2 Other	7.50	7.50	
RATE						SURVEYS: MD INC' AZ'			3	TOTALS	24.00	32.50		
LINER			6.0"			0 0 0			Potassium Chloride (bulk)	4	NON PRODUCTIVE	6.50	439.50	
#2: NAT 8-P-80			STROKE			9.0"			Sodium Sulphite	6	PRODUCTS: USED REC ON SITE			
RATE									Xan-Bore	19	DIESEL (L)	3650		44100
LINER			6.0"								BARITE (T)	-0.7		26.0
#3: IDECO T-800			STROKE			8.5"					DRILL WATER	14000		44000
RATE											DAILY MUD COSTS \$6,793.50			
LINER			6.0"								CUM. MUD COSTS \$11,585.74			

HOURLY OPERATIONS SUMMARY 0000 to 2400			
From	To	IADC Code	Description
0:00	3:00	[12.2]	NPT- Waited on Halliburton cementers to reset hours.
3:00	4:30	[12.2]	Rigged up to cement, PJSM, mixed and pumped 3.5 bbls 15.8 ppg cement plug through stinger inside casing and spotted on bottom.
4:30	6:30	[13]	Waited on cement. Laid out stinger.
6:30	7:30	[12.2]	PJSM, with Halliburton and crew. Mix and pumped 17bbls of 15.8ppg cement slurry at 1.5bpm, good returns to cellar floor after 8bbls, flushed lines.
7:30	10:30	[13]	WOC checked samples hard after 3hrs, Prepare 17 1/2" BHA.
10:30	11:30	[21.2]	PJSM, PTW. Cut 20" conductor off 200mm below matting and lay out cut-off.
11:30	12:00	[21.2]	Held PTSM discussed PTW system, hazard identification, working at heights identifying correct fall restraints.
12:00	18:00	[21.2]	Nipped up 20" riser and flow line.
18:00	18:30	[8]	NPT- Repaired elevator hydraulic hose.
18:30	20:30	[6.3]	Broke out 24" bit. Made up 17 1/2" bit. Installed 17 1/2" stabiliser and Totco ring.
20:30	21:00	[7]	Serviced rig.
21:00	21:45	[8]	NPT- Repaired mud pump #1
21:45	23:30	[8]	NPT- Safety shutdown due to incident - injured finger.
23:30	0:00	[8]	NPT- Repaired mud pump #1

HOURLY OPERATIONS SUMMARY 0000 to 0600 on 23/08/18			
From	To	Description	
0:00	6:00	Repaired mud pump #1	

POB-37: Ensign - 19, CTP - 4, OICS - 3, Halliburton - 4, Pason - 1, OCS - 3, SiteMed - 1, RMN - 1, ERMS - 1		10,296 Hrs
RECEIVED: Drill water		DISPATCHED: Tango mix pump
MAXIMUM GAS:	BACKGROUND GAS:	CONNECTION GAS:
SUPERVISOR: Kev Dau / Mark Booth / Dave York	GEOLOGIST: Phil Allen / Peter Morris	RIG MGR: Gary Hender / Adam Delaine

POB-36: Ensign - 20, CTP - 4, OICS - 3, Halliburton - 4, OCS - 2, SiteMed - 1, RMN - 1, WFT - 1				10,728 Hrs
RECEIVED: Drill water. 1 X 9 5/8" Fluid Hammer, 19 X pallets mud chemical		DISPATCHED:		
MAXIMUM GAS:	BACKGROUND GAS:	CONNECTION GAS:	TRIP GAS:	
SUPERVISOR: Kev Dau / Dave York	GEOLOGIST: Phil Allen / Peter Morris	RIG MGR: Simon Luke Bryant / Adam Delaine		

DAILY DRILLING REPORT

DAILY DRILLING REPORT

POB-35: Ensign - 20, CTP - 4, OICS - 3, Halliburton - 4, OCS - 1, Site Med - 1, RMN - 1, WFT - 1				12,432 Hrs
RECEIVED: 35 x 36# 9 5/8" CSG			DISPATCHED: 8 x 13 3/8" CSG	
MAXIMUM GAS:		BACKGROUND GAS:		CONNECTION GAS:
SUPERVISOR: Kev Dau / Dave York		GEOLOGIST: Phil Allen / Peter Morris		RIG MGR: Simon Luke Bryant / Alan Delaine



DAILY DRILLING REPORT

REPORT # 00828/08/2018

SPUD: 21/08/2018DAY # 054

WELL

Palm Valley 13

24:00 MD/ TVD

527m BRT

DRILLED

33m

AFE DAYS / COST

46

\$13,333,080

RIG

ENSIGN 932

FORMATION

Mereenie Sandstone

DAILY COST

\$104,747

CUM.

\$5,977,056

DAILY OP'S SUMMARY

Drilled 12 1/4" hole from 494m to 518m. Tripped out of the hole for downhole pressure loss. Laid down damaged motor and made up 12 1/4" BHA.

Tripped in the hole and drilled 12 1/4" hole from 518m to 527m.

FORECAST OP'S

Drill 12 1/4" hole.

LAST CASING

340mm

13.375"

SET AT

249.0m

MAASP

BOP TEST

NIL

DUE

LAST L.T.I.

437 Days

SAFETY

1. PTSM: Review BBS cards. Servicing handling equipment. Housekeeping.

2. PTSM: Upcoming operations. Laying down and picking up BHA. Threaded connection care.

WEATHER: Day

Fine

Night

Fine

BIT INFORMATION				BHA # 4		MUD PROPERTIES		OPERATION	HRS	CUM
WOB(Klb)	5-23	JET V(fps)	219	TOOL	LENGTH	Mud Type	KCl Gel Polymer	1 Move/ Rig Up/Down		1119.50
RPM	100	H S I	1.70	Bit - Tri Cone	0.33	Depth (m)	522	2 Drilling	5.50	71.50
BIT NUMBER	4	3		Bit Sub	1.20	Density (ppg)	9.10	3 Wash / Ream	0.50	2.00
Size (inches)	12.25	12.25		2x Drill Collar- 9 1/2"	18.63	ECD (ppg)		4 Coring		
Make	Smith	Reed		Crossover Sub	1.09	Temp (° C)	41	5 Circ & Condition		5.75
Type	GFI288 ODVPS	TK66		Stabiliser- UG 1/4"	1.35	Viscosity (sec)	40	6 Tripping- Bit / Casing		7.25
IADC Code				Drill Collar- 8"	9.47	PV / YP (cp/lb)	9 / 21	6.1 Tripping- Other	8.00	8.00
Serial Number	RJ4049	E232507		Crossover Sub	0.53	Gels (10s/m)	6 / 9	6.2 Wiper Trip		
T.F.A. (in)	1.286	1.335		16x Drill Collar- 6 1/2"	148.77	API Filt. (cc)	8.0	6.3 Handle BHA	6.50	16.50
Depth In (m)	518	249		Crossover Sub	0.79	Sand (% Vol)	tr	7 Service / Slip Line	2.00	4.25
Depth Out (m)	IN	518		Drilling Jars	9.62	KCl (%)	3.3	8 Repairs	1.00	14.00
Total Meters	9	269		3x Drill Collar- 6 1/2"	27.41	pH (strip)	9.5	10 Survey		1.75
Hours IADC OnBtm	2 1.4	25.8 21.2		Crossover Sub	0.81	LGS %/Vol	4.1	11 Logging		
ROP IADC OnBtm	4.5 6.4	10.4 12.7		6x Heavy Weight DP	56.40	Chlorides (ppm)	13000	12 RU for Csg / Cmt		11.25
Condition Out	2 2 WT A X 0 CT DMF					Surface Vol. (Bbls)	1129	12.1 Run Casing		7.00
FLOW DATA				BHA LENGTH	276.40	Hole Vol.	226	12.2 Cementing		9.25
RATE (gpm/lps)	878	55.4		BHA WEIGHT	66.29	Mud Mixed	180	13 Wait on Cement		9.50
AV - DP (fpm/mpm)	160	49		STRING WT	77.49	New Hole Drilled	15.8	14 BOP's / Wellhead		
AV - DC (fpm/mpm)	360	110		HOOK LOAD	95.00	Downhole Losses		15 Test BOP / Formation		
SPP (psi/atm)	1665	113		WT BELOW JARS	55.96	Surface Losses	213	19 Tight hole / Fishing		
SPP (Calculated)				HOURS: JAR MTR	28.0	CHEMICAL USAGE		21 Completion		
PUMP DATA				DRAG: UP DOWN	3 2	AMC Biocide G	2	21.1 Well Control		
#1: NAT 8-P-80	STROKE	9.0"		TORQUE: ON OFF	3600 1500	AMCPacLV	6	21.2 Other	0.50	8.50
RATE	93			SURVEYS: MD INC' AZ'		Caustic Soda	2	TOTALS	24.00	176.50
LINER	6.0"			409 1 0		Potassium Chloride	45	NON PRODUCTIVE	16.50	466.00
#2: NAT 8-P-80	STROKE	9.0"				Soda Ash	1	PRODUCTS: USED REC ON SITE		
RATE	93					Sodium Sulphite	1	DIESEL (L)	7250	22000 61850
LINER	6.0"					Xan-Bore	8	BARITE (T)		26.0
#3: IDECO T-800	STROKE	8.5"						DRILL WATER	6000	45000
RATE	93							DAILY MUD COSTS \$2,360.42		
LINER	6.0"							CUM. MUD COSTS \$28,362.97		

HOURLY OPERATIONS SUMMARY 0000 to 2400

From	To	[IADC Code] Description
0:00	1:00	[2] Drilled 12 1/4" hole from 494m to 503m with 35klbs WOB, 860GPM, 2100psi, 35 surface RPM, 6400 ft.lbs Trq. Average ROP: 9m/hr
1:00	2:00	[8] NPT- Repaired rig - reset Pason after outage.
2:00	4:30	[2] Drilled 12 1/4" hole from 503m to 518m with 35klbs WOB, 860GPM, 2100psi, 35 surface RPM, 4800 ft.lbs Trq. Average ROP: 6 m/hr
4:30	5:00	[21.2] NPT- Observe pressure drop of 450psi in both standpipe and differential coinciding with 18klbs WOB applied to bit while attempting to drill.
5:00	8:30	[6.1] NPT- Trip out of hole 518m to 52m. due to down hole pressure loss.
8:30	10:30	[6.3] NPT- Broke out and inspected leaking 8" drill collar thread. Lay out 3 X 8" drill collars with damaged threads.
10:30	11:15	[6.3] NPT- Rig up HT100 tongs and check tong load cell.
11:15	12:00	[6.3] NPT- Changed out 6 5/8 X 7 5/8 crossover sub and made up 8" drill collar.
12:00	14:30	[6.3] NPT- Pulled out of hole BHA from 52m to surface inspecting all joints and lay down damaged Fluid Hammer.
14:30	16:30	[7] Conducted mast inspection, tightened torque track bolts, reseal riser.
16:30	17:00	[6.3] NPT- Prepared 12 1/4" BHA components.
17:00	21:30	[6.1] NPT- Made up 12 1/4" BHA and tripped in to 510m.
21:30	22:00	[3] NPT- Washed to bottom and feathered on and off bottom, pumping hi-vis sweep. No junk detected.
22:00	0:00	[2] Drilled 12 1/4" hole from 518m to 527m. 23klbs WOB, 100RPM, 860GPM at 1665psi., 3600ft/lbs Torque. Av ROP: 4.5m/hr.

HOURLY OPERATIONS SUMMARY 0000 to 0600 on 29/08/18

0:00	4:45	Drilled 12 1/4" hole from 527m to 547m. 23klbs WOB, 100RPM, 840GPM at 1610psi., 3600ft/lbs Torque. Av ROP: 4.4m/hr.
4:45	6:00	Repaired rig - malfunctioning Top Drive rotary function.

POB-36: Ensign - 20, CTP - 4, OICS - 3, Halliburton - 5, OCS - 1, Site Med - 1, RMN - 1, WFT - 1

RECEIVED: 55 x 36# 9 5/8" casing

DISPATCHED:

12,864 Hrs

MAXIMUM GAS:

BACKGROUND GAS:

CONNECTION GAS:

TRIP GAS:

SUPERVISOR: Kev Dau / Dave York

GEOLOGIST: Phil Allen / Peter Morris

RIG MGR: Simon Luke Bryant / Alan Delaine



DAILY DRILLING REPORT

REPORT # 00929/08/2018

SPUD: 21/08/2018DAY # 055

WELL

Palm Valley 13

24:00 MD/ TVD

622m BRT

DRILLED

95m

AFE DAYS / COST

46

\$13,467,756

RIG

ENSIGN 932

FORMATION

Mereenie Sandstone

DAILY COST

\$86,602

CUM.

\$6,063,659

DAILY OP'S SUMMARY

Drilled 12 1/4" hole from 527m to 622m with rig repair, wireline survey and rig service.

FORECAST OP'S

Drill 12 1/4" hole.

LAST CASING

340mm

13.375"

SET AT

249.0m

MAASP

BOP TEST

NIL

DUE

LAST L.T.I.

438 Days

SAFETY

1. PTSM: Review BBS cards. Care of threaded connections. Prep 9 5/8" casing.

2. PTSM: Housekeeping. Wireline survey JSA. Rig service.

WEATHER: Day

Fine

Night

Fine

BIT INFORMATION				BHA # 4		MUD PROPERTIES			OPERATION		HRS	CUM
WOB(Klb)	23-38	JET V(fps)	184	TOOL	LENGTH	Mud Type	KCI Gel Polymer		1 Move/ Rig Up/Down			1119.50
RPM	100	H S I	1.00	Bit - Tri Cone	0.33	Depth (m)	564	604	2 Drilling	21.50		93.00
BIT NUMBER		4	3	Bit Sub	1.20	Density (ppg)	9.05	9.05	3 Wash / Ream			2.00
Size (inches)		12.25	12.25	2x Drill Collar- 9 1/2"	18.63	ECD (ppg)	9.21	9.21	4 Coring			
Make		Smith	Reed	Crossover Sub	1.09	Temp (° C)	51	55	5 Circ & Condition			5.75
Type		GF1288 ODVPS	TK66	Stabiliser- UG 1/4"	1.35	Viscosity (sec)	41	40	6 Tripping- Bit / Casing			7.25
IADC Code				Drill Collar- 8"	9.47	PV / YP (cp/lb)	9 / 23	8 / 22	6.1 Tripping- Other			8.00
Serial Number		RJ4049	E232507	Crossover Sub	0.53	Gels (10s/m)	9 / 10	6 / 10	6.2 Wiper Trip			
T.F.A. (in)		1.286	1.335	16x Drill Collar- 6 1/2"	148.77	API Filt. (cc)	7.2	7.2	6.3 Handle BHA			16.50
Depth In (m)		518	249	Crossover Sub	0.79	Sand (% Vol)	tr	tr	7 Service / Slip Line	0.50		4.75
Depth Out (m)		IN	518	Drilling Jars	9.62	KCI (%)	3.5	3.3	8 Repairs	2.00		16.00
Total Meters		104	269	3x Drill Collar- 6 1/2"	27.41	pH (strip)	9.5	9.5	10 Survey			1.75
Hours IADC OnBtm		23.5 21.4	25.8 21.2	Crossover Sub	0.81	LGS %/Vol	3.5	3.8	11 Logging			
ROP IADC OnBtm		4.4 4.9	10.4 12.7	6x Heavy Weight DP	56.40	Chlorides (ppm)	13000	13500	12 RU for Csg / Cmt			11.25
Condition Out		2 2 WT A X 0 CT DMF				Surface Vol. (Bbls)		1161	12.1 Run Casing			7.00
FLOW DATA				BHA LENGTH		Hole Vol.		270	12.2 Cementing			9.25
RATE (gpm/lps)		736	46.4	BHA WEIGHT		Mud Mixed		323	13 Wait on Cement			9.50
AV - DP (fpm/mpm)		135	41	STRING WT		New Hole Drilled		45.4	14 BOP's / Wellhead			
AV - DC (fpm/mpm)		302	92	HOOK LOAD		Downhole Losses			15 Test BOP / Formation			
SPP (psi/atm)		1160	79	WT BELOW JARS		Surface Losses		246	19 Tight hole / Fishing			
SPP (Calculated)		1010	69	HOURS: JAR MTR		CHEMICAL USAGE			21 Completion			
#1: NAT 8-P-80		STROKE	9.0"	DRAG: UP DOWN		AMC Biocide G		1	21.1 Well Control			
RATE		78		TORQUE: ON OFF		AMCPacLV		8	21.2 Other			8.50
LINER		6.0"		SURVEYS: MD INC' AZ'		Caustic Soda		1	TOTALS	24.00		200.50
#2: NAT 8-P-80		STROKE	9.0"	409 1 0		Soda Ash		3	NON PRODUCTIVE	2.00		468.00
RATE		78		603 0.5 0		Sodium Sulphite		5	PRODUCTS: USED REC ON SITE			
LINER		6.0"				Xan-Bore		9	DIESEL (L)	8150		53700
#3: IDECO T-800		STROKE	8.5"						BARITE (T)			26.0
RATE		78							DRILL WATER	5500		39500
LINER		6.0"							DAILY MUD COSTS			\$1,840.23
									CUM. MUD COSTS			\$30,203.20

HOURLY OPERATIONS SUMMARY 0000 to 2400

From	To	[IADC Code] Description
0:00	4:45	[2] Drilled 12 1/4" hole from 527m to 547m. 23klbs WOB, 100RPM, 840GPM at 1610psi., 3600ft/lbs Torque. Av ROP: 4.4m/hr.
4:45	6:15	[8] NPT- Repaired rig - malfunctioning Top Drive rotary function.
6:15	12:00	[2] Drilled 12 1/4" hole from 527m to 547m. 23 - 35klbs WOB, 100RPM, 840 - 720GPM at 1610 - 1200psi., 4000 - 6000ft/lbs Torque. Av ROP: 4.1m/hr.
12:00	19:30	[2] Drilled 12 1/4" hole from 547m to 605m. 35klbs WOB, 100RPM, 720GPM at 1150 psi., 5000ft/lbs Torque. Av ROP: 6.4m/hr.
19:30	20:00	[7] Serviced rig.
20:00	23:30	[2] Drilled 12 1/4" hole from 605m to 620m. 35klbs WOB, 100RPM, 720GPM at 1160 psi., 5000ft/lbs Torque. Av ROP: 5 m/hr.
23:00	23:30	[8] NPT- Repaired rig - Pason dropped out - comms failure.
23:30	0:00	[2] Drilled 12 1/4" hole from 620m to 622m. 35klbs WOB, 100RPM, 720GPM at 1160 psi., 5000ft/lbs Torque. Av ROP: 4m/hr.

HOURLY OPERATIONS SUMMARY 0000 to 0600 on 30/08/18

0:00	6:00	Drilled 12 1/4" hole from 622m to 647m. 35-38klbs WOB, 100RPM, 720GPM at 1160 psi., 5000ft/lbs Torque. Av ROP: 4.2m/hr.
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POB-37: Ensign - 20, CTP - 4, OICS - 3, Halliburton - 4, OCS - 1, Site Med - 2, RMN - 2, WFT - 1

13,308 Hrs

RECEIVED:

DISPATCHED:

MAXIMUM GAS:

BACKGROUND GAS:

CONNECTION GAS:

TRIP GAS:

SUPERVISOR: Andy Phillis / Dave York

GEOLOGIST: Phil Allen / Peter Morris

RIG MGR: Simon Luke Bryant / Alan Delaine

WELL	Palm Valley 13	24:00 MD/ TVD	701m BRT	DRILLED	79m	AFE DAYS / COST	46	\$13,467,756			
RIG	ENSIGN 932	FORMATION	Mereenie Sandstone	DAILY COST	\$85,816	CUM.	\$6,149,475				
DAILY OP'S SUMMARY	Drilled 12 1/4" hole from 622m to 701m with a rig service. Circulated bottoms up and tripped out of the hole for a bit change to 22m.										
FORECAST OP'S	Drill 12 1/4" hole.										
LAST CASING	340mm	13.375"	SET AT	249.0m		MAASP		BOP TEST	NIL	DUE	
LAST L.T.I. 439 Days	SAFETY	1. PTSM: Reviewed BBS cards. MSDS sheets. Correct PPE.					WEATHER: Day		Fine		
		2. PTSM: Drill ahead. Bit trip coming up. Communication.					Night		Fine		

BIT INFORMATION				BHA # 4		MUD PROPERTIES			OPERATION		HRS	CUM			
WOB(Klb)	35-38	JET V(fps)	186	TOOL		LENGTH		Mud Type	KCl Gel Polymer		1 Move/ Rig Up/Down		1119.50		
RPM	105	H S I	1.04	Bit - Tri Cone		0.33		Depth (m)	664	697	2 Drilling	20.50	113.50		
BIT NUMBER		4	3	Bit Sub		1.20		Density (ppg)	9.10	9.10	3 Wash / Ream		2.00		
Size (inches)		12.25	12.25	2x Drill Collar- 9 1/2"		18.63		ECD (ppg)	9.23	9.23	4 Coring				
Make		Smith	Reed	Crossover Sub		1.09		Temp (* C)	49	47	5 Circ & Condition	0.50	6.25		
Type		GF288 ODVPS	TK66	Stabiliser- UG 1/4"		1.35		Viscosity (sec)	41	41	6 Tripping- Bit / Casing	2.50	9.75		
IADC Code				Drill Collar- 8"		9.47		PV / YP (cp/lb)	9 / 21	8 / 22	6.1 Tripping- Other		8.00		
Serial Number		RJ4049	E232507	Crossover Sub		0.53		Gels (10s/m)	7 / 10	7 / 10	6.2 Wiper Trip				
T.F.A. (in)		1.286	1.335	16x Drill Collar- 6 1/2"		148.77		API Filt. (cc)	7.2	7.3	6.3 Handle BHA		16.50		
Depth In (m)		518	249	Crossover Sub		0.79		Sand (% Vol)	0.1	0.1	7 Service / Slip Line	0.50	5.25		
Depth Out (m)		701	518	Drilling Jars		9.62		KCl (%)	3.0	3.0	8 Repairs		16.00		
Total Meters		183	269	3x Drill Collar- 6 1/2"		27.41		pH (strip)	9.5	9.5	10 Survey		1.75		
Hours IADC OnBtm		44 39.9	25.8 21.2	Crossover Sub		0.81		LGS %/Vol	4.3	4.3	11 Logging				
ROP IADC OnBtm		4.2 4.6	10.4 12.7	6x Heavy Weight DP		56.40		Chlorides (ppm)	13250	13000	12 RU for Csg / Cmt		11.25		
Condition Out		6 8 BT G E 2 WT HR						Surface Vol. (Bbls)		1275	12.1 Run Casing		7.00		
FLOW DATA				BHA LENGTH		276.40		Hole Vol.		306	12.2 Cementing		9.25		
RATE (gpm/lps)		746	47.0	BHA WEIGHT		86.29		Mud Mixed		187	13 Wait on Cement		9.50		
AV - DP (fpm/mpm)		136	42	STRING WT		55.26		New Hole Drilled		37.8	14 BOP's / Wellhead				
AV - DC (fpm/mpm)		306	93	HOOK LOAD		104.00		Downhole Losses			15 Test BOP / Formation				
SPP (psi/atm)		1260	86	WT BELOW JARS		55.96		Surface Losses		35	19 Tight hole / Fishing				
SPP (Calculated)		1120	76	HOURS: JAR MTR		70.0		CHEMICAL USAGE			21 Completion				
PUMP DATA				DRAG: UP DOWN		2	2	AMC Biocide G		2	21.1 Well Control				
#1: NAT 8-P-80		STROKE	9.0"	TORQUE: ON OFF		5000	1500	Caustic Soda		2	21.2 Other		8.50		
RATE		79		SURVEYS: MD INC AZ				Potassium Chloride		19	TOTALS		24.00	224.50	
LINER		6.0"		603 0.5 0				Sodium Sulphite		6	NON PRODUCTIVE			468.00	
#2: NAT 8-P-80		STROKE	9.0"					Xan-Bore		2	PRODUCTS: USED			REC ON SITE	
RATE		79									DIESEL (L)		8400		45300
LINER		6.0"									BARITE (T)				26.0
#3: IDECO T-800		STROKE	8.5"								DRILL WATER		6000		33500
RATE		79									DAILY MUD COSTS		\$1,054.00		
LINER		6.0"									CUM. MUD COSTS		\$31,257.20		

[illegible]

HOURLY OPERATIONS SUMMARY 0000 to 0600 on 31/08/18		
0:00	0:30	Break Bit #4 .Make up Bit#5
0:30	4:15	RIH from Surface to 548m .Inspect tool faces .Lay down 1 x HWDP-Galled face
4:15	6:00	Wash/Ream from 548m to 622m .Intermittent Hi -torque .

POB-35: Ensign - 20, CTP - 4, OICS - 3, Halliburton - 4, OCS - 1, Site Med - 1, RMN - 1, WFT - 1				13,728 Hrs
RECEIVED:		DISPATCHED:		
MAXIMUM GAS:		BACKGROUND GAS:		CONNECTION GAS:
SUPERVISOR: Andy Phillis / Dave York		GEOLOGIST: Phil Allen / Peter Morris		RIG MGR: Simon Luke Bryant / Adam Delaine

POB-38: Ensign - 20, CTP - 4, OICS - 3, Halliburton - 4, OCS - 2, Site Med - 1, RMN - 1, WFT - 3				14,628 Hrs
RECEIVED: 2 X 6 1/2" Fluid Hammer, 1 X 6 1/2" jars.			DISPATCHED:	
MAXIMUM GAS:		BACKGROUND GAS:		CONNECTION GAS:
TRIP GAS:				
SUPERVISOR: Andy Phillis / Dave York		GEOLOGIST: Phil Allen / Peter Morris		RIG MGR: Simon Luke Bryant / Adam Delaine



DAILY DRILLING REPORT

REPORT # 013

2/09/2018

SPUD: 21/08/2018

DAY # 059

WELL

Palm Valley 13

24:00 MD/ TVD

758m BRT

DRILLED

AFE DAYS / COST

46

\$13,467,756

RIG

ENSIGN 932

FORMATION

Mereenie Sandstone

DAILY COST

\$82,960

CUM.

\$6,474,132

DAILY OP'S SUMMARY

Tripped out of hole to find twisted off crossover sub. Waited on arrival of fishing tools, while preparing drill collars for inspection. Upon delivery, prepared fishing tools for use.

FORECAST OP'S

Wash over fish and attempt engage with overshot.

LAST CASING

340mm

13.375"

SET AT

249.0m

MAASP

BOP TEST

NIL

DUE

LAST L.T.I.

442 Days

SAFETY

1. Review BBS cards. Tripping pipe. Manual handling. New hand on floor.

2. PTSM: Weekly Safety Meeting. New heat stress policy. Fishing ops.

WEATHER: Day

Fine

Night

Fine

BIT INFORMATION					MUD PROPERTIES		OPERATION	HRS	CUM
WOB(Klb)	JET V(fps)		TOOL	LENGTH	Mud Type	KCl Gel Polymer	1 Move/ Rig Up/Down		1119.50
RPM	H S I				Depth (m)	758	2 Drilling		137.75
BIT NUMBER	7	RR6			Density (ppg)	9.25	3 Wash / Ream		6.50
Size (inches)	12.25	12.25			ECD (ppg)		4 Coring		
Make	Stealth	Reed			Temp (° C)		5 Circ & Condition		9.25
Type	YCJXCN	R50PDH2			Viscosity (sec)	41	6 Tripping- Bit / Casing		22.75
IADC Code	437	637			PV / YP (cp/lb)	7 / 23	6.1 Tripping- Other	2.50	11.00
Serial Number	2017	AT3343			Gels (10s/m)	6 / 8	6.2 Wiper Trip		
T.F.A. (in)	0.920	4.059			API Filt. (cc)	6.8	6.3 Handle BHA	8.00	25.75
Depth In (m)	738	703			Sand (% Vol)	0.1	7 Service / Slip Line		6.75
Depth Out (m)	IN	738			KCl (%)	3.2	8 Repairs		16.00
Total Meters	20	35			pH (strip)	9.5	10 Survey		1.75
Hours IADC OnBtm	6.5 5.8	13.3 11.8			LGS %/Vol	5.4	11 Logging		
ROP IADC OnBtm	3.1 3.4	2.6 3			Chlorides (ppm)	15000	12 RU for Csg / Cmt		11.25
Condition Out	1 1 FC A E 2 FC HP				Surface Vol. (Bbls)	1077	12.1 Run Casing		7.00
FLOW DATA			BHA LENGTH		Hole Vol.	369	12.2 Cementing		9.25
RATE (gpm/lps)			BHA WEIGHT		Mud Mixed	20	13 Wait on Cement		9.50
AV - DP (fpm/mpm)			STRING WT		New Hole Drilled		14 BOP's / Wellhead		
AV - DC (fpm/mpm)			HOOK LOAD		Downhole Losses		15 Test BOP / Formation		
SPP (psi/atm)			WT BELOW JARS		Surface Losses	61	19 Tight hole / Fishing		
SPP (Calculated)			HOURS: JAR MTR	94.0	CHEMICAL USAGE		21 Completion		
PUMP DATA			DRAG: UP DOWN		AMC Biocide G	1	21.1 Well Control		
#1: NAT 8-P-80	STROKE	9.0"	TORQUE: ON OFF		Caustic Soda	1	21.2 Other	13.50	22.00
RATE			SURVEYS: MD INC° AZ°		Sodium Sulphite	2	TOTALS	24.00	296.50
LINER	6.0"		603 0.5 0				NON PRODUCTIVE	24.00	492.50
#2: NAT 8-P-80	STROKE	9.0"					PRODUCTS: USED REC ON SITE		
RATE							DIESEL (L)	9132	29982 54850
LINER	6.0"						BARITE (T)		26.0
#3: IDECO T-800	STROKE	8.5"					DRILL WATER	6000	32000
RATE							DAILY MUD COSTS \$197.97		
LINER	6.0"						CUM. MUD COSTS \$32,598.76		

HOURLY OPERATIONS SUMMARY 0000 to 2400

From	To	[IADC Code] Description
0:00	2:30	[6.1] NPT- Tripped out of hole to inspect BHA. Found twisted off crossover sub. Top of fish at 725.76m.
2:30	3:00	[6.3] NPT- Laid out 6 1/2" drill collar with partial twisted off crossover sub made up.
3:00	11:00	[21.2] NPT- Assessed fishing gear on location. Waited on delivery of fishing equipment.
11:00	14:30	[6.3] NPT- Tripped in hole with 6 1/2" drill collars, inspecting connections.
14:30	16:00	[6.3] NPT- Laid out excess drill collars, not required for fishing, to be inspected.
16:00	21:30	[21.2] NPT- Waited on delivery of fishing tools.
21:30	0:00	[6.3] NPT- Assessed options with available fishing tools. Cleaned all rust scale from threads and interior of overshot.

HOURLY OPERATIONS SUMMARY 0000 to 0600 on 03/09/18

0:00	1:00	Cleaned rust scale from interior of overshot and dressed overshot with 7 1/2" basket grapple and 7 1/2" mill control (ID = 7 5/8") with 7 1/2" pack off.
1:00	2:00	Made up fishing BHA.
2:00	6:00	Tripped in with fishing assembly to 705m., picking up 14 singles of drill pipe.

POB-38: Ensign - 20, CTP - 4, OICS - 3, Halliburton - 4, OCS - 2, Site Med - 1, RMN - 1, WFT - 3

15,084 Hrs

RECEIVED:

DISPATCHED:

MAXIMUM GAS:

BACKGROUND GAS:

CONNECTION GAS:

TRIP GAS:

SUPERVISOR: Andy Phillis / Dave York

GEOLOGIST: Phil Allen / Peter Morris

RIG MGR: Simon Luke Bryant / Adam Delaine

DAILY DRILLING REPORT

POB-39: Ensign - 20, CTP - 4, OICS - 3, Halliburton - 5, OCS - 2, Site Med - 1, RMN - 1, WFT - 3				16,008 Hrs
RECEIVED: 1 X Tango overshoot, extension bowl, grapple and control. 1 X Tango crossover sub NC46 X 6 5/8 Reg.		DISPATCHED:		
MAXIMUM GAS:	BACKGROUND GAS:	CONNECTION GAS:	TRIP GAS:	
SUPERVISOR: Andy Phillis / Dave York	GEOLOGIST: Phil Allen / Peter Morris	RIG MGR: Simon Luke Bryant / Adam Delaine		

DAILY DRILLING REPORT



DAILY DRILLING REPORT

REPORT # 018

7/09/2018

SPUD: 21/08/2018

DAY # 064

WELL	Palm Valley 13	24:00 MD/ TVD	955m BRT	DRILLED	197m	AFE DAYS / COST	46	\$13,467,756
RIG	ENSIGN 932	FORMATION	Mereenie Sandstone	DAILY COST	\$91,230	CUM.	\$7,001,633	

DAILY OP'S SUMMARY

Washed and reamed 557m to 758m. Conditioned mud. Drilled 12 1/4" hole from 758m to 955m., with rig service.

FORECAST OP'S

Drill 12 1/4" hole

LAST CASING	340mm	13.375"	SET AT	249.0m	MAASP	BOP TEST	NIL	DUE
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LAST L.T.I. 447 Days

SAFETY

1. PTSM: Reaming in the hole. Housekeeping. Handling tubulars.
2. PTSM: Make up doubles for drilling. Conditioning mud. Housekeeping.

WEATHER: Day

Fine

Night

Fine

HOURLY OPERATIONS SUMMARY 0000 to 2400

From	To	[IADC Code] Description
0:00	3:30	[3] Reamed and washed intermittent high torque intervals from 557m to 750m.
3:30	3:45	[3] Washed down from 750m to 758m and tagged.
3:45	4:00	[5] Pumped 20bbls Hi-Vis sweep off bottom and displaced out of string.
4:00	5:00	[2] Bedded in bit. Drilled 12 1/4" hole from 758m to 764m. 30klbs WOB, 75 - 80RPM, 750GPM at 1650 psi., 6300ft/lbs Torque. Av ROP: 8m/hr.
5:00	6:00	[5] Circulate and condition mud - mud returns exhibiting very low viscosity and aeration.
6:00	9:30	[2] Drilled 12 1/4" hole from 764m to 807m. 30klbs WOB, 90RPM, 760GPM at 1800 psi., 12000ft/lbs Torque. Av ROP: 13.4m/hr.
9:30	10:00	[7] Serviced Rig
10:00	10:30	[10] Run Wireline Survey @ 797m.1.0 Degrees
10:30	18:00	[2] Drilled 12 1/4" hole from 807m to 915m. 35klbs WOB, 90RPM, 760GPM at 1940 psi., 10000ft/lbs Torque. Av ROP: 16.1 m/hr.
18:00	0:00	[2] Drilled 12 1/4" hole from 915m to 955m. 35 - 45klbs WOB, 80 - 90RPM, 760GPM at 1940 psi., 6500 - 10000ft/lbs Torque. Av ROP: 6.7 m/hr.

HOURLY OPERATIONS SUMMARY 0000 to 0600 on 08/09/18

0:00	4:00	Drilled 12 1/4" hole from 955m to 971m. 45klbs WOB, 80 RPM, 760GPM at 1940 psi., 6500 ft/lbs Torque. Av ROP: 4 m/hr.
4:00	4:45	Investigate 300psi standpipe pressure loss. Diagnose fault in #1 mud pump.
4:45	6:00	Drilled 12 1/4" hole from 971m to 974m. 45klbs WOB, 85 RPM, 700GPM at 1740 psi., 7000 ft/lbs Torque. Av ROP: 2.4 m/hr.

POB-38: Ensign - 19, CTP - 4, OICS - 3, Halliburton - 4, OCS - 2, Site Med - 1, RMN - 1, WFT - 4

17,436 Hrs

RECEIVED:

DISPATCHED:



DAILY DRILLING REPORT

REPORT # 0198/09/2018

SPUD: 21/08/2018DAY # 065

WELL

Palm Valley 13

24:00 MD/ TVD

1014m BRT

DRILLED

59m

AFE DAYS / COST

46

\$13,467,756

RIG

ENSIGN 932

FORMATION

Carmichael Sst.

DAILY COST

\$88,768

CUM.

\$7,090,401

DAILY OP'S SUMMARY

Drilled 12 1/4" hole from 955m to 1014m with rig service and survey.

FORECAST OP'S

Drill 12 1/4" hole.

LAST CASING

340mm

13.375"

SET AT

249.0m

MAASP

BOP TEST

NIL

DUE

LAST L.T.I.

448 Days

SAFETY

1. PTSM: Reviewed BBS cards. Drilling ahead. Upcoming ops.

2. PTSM: Reviewed BBS cards. Drilling ahead. Training new personnel.

WEATHER: Day

Fine

Night

Fine

BIT INFORMATION				BHA # 5		MUD PROPERTIES			OPERATION	HRS	CUM			
WOB(Klb)	40-45	JET V(fps)	192	TOOL	LENGTH	Mud Type	KCI Gel Polymer		1 Move/ Rig Up/Down		1119.50			
RPM	80	H S I	1.12	Bit - PDC	0.25	Depth (m)	980	1003	2 Drilling	22.25	178.00			
BIT NUMBER		8	7	Near Bit Stabiliser	1.79	Density (ppg)	9.35	9.30	3 Wash / Ream		12.75			
Size (inches)		12.25	12.25	Float Sub	0.56	ECD (ppg)	9.46	9.41	4 Coring					
Make		HTC	Stealth	2x Drill Collar- 8"	18.62	Temp (° C)	48	50	5 Circ & Condition		10.50			
Type		KMX524T	YCJXCN	Stabiliser	1.34	Viscosity (sec)	38	39	6 Tripping- Bit / Casing		23.75			
IADC Code			437	2x Drill Collar- 8"	18.81	PV / YP (cp/lb)	8 / 20	9 / 22	6.1 Tripping- Other		19.50			
Serial Number		5281617	2017	Crossover Sub	0.79	Gels (10s/m)	6 / 8	6 / 8	6.2 Wiper Trip					
T.F.A. (in)		1.230	0.920	16x Drill Collar- 6 1/2"	149.89	API Filt. (cc)	9.8	7.8	6.3 Handle BHA		37.00			
Depth In (m)		758	738	Crossover Sub	0.80	Sand (% Vol)	tr	tr	7 Service / Slip Line	0.50	7.75			
Depth Out (m)		IN	758	Drilling Jars	9.60	KCI (%)	2.8	2.8	8 Repairs		62.00			
Total Meters		256	20	Crossover Sub	0.58	pH (strip)	9.5	9.5	10 Survey	0.50	2.75			
Hours IADC OnBtm		40.3 36.7	6.5 5.8	Drill Collar- 6 1/2"	9.39	LGS %/Vol	6.3	6.0	11 Logging					
ROP IADC OnBtm		6.4 7	3.1 3.4	Crossover Sub	0.80	Chlorides (ppm)	14750	15000	12 RU for Csg / Cmt		11.25			
Condition Out		1 2 BT M E 0 NO TW			8x Heavy Weight DP	75.15	Surface Vol. (Bbls)		1233		12.1 Run Casing	7.00		
FLOW DATA				BHA LENGTH		288.37		Hole Vol.		450	12.2 Cementing	9.25		
RATE (gpm/lps)		736	46.4	BHA WEIGHT		67.10		Mud Mixed		197	13 Wait on Cement	9.50		
AV - DP (fpm/mpm)		135	41	STRING WT		99.37		New Hole Drilled		28.2	14 BOP's / Wellhead			
AV - DC (fpm/mpm)		223	68	HOOK LOAD		119.00		Downhole Losses			15 Test BOP / Formation			
SPP (psi/atm)		1900	129	WT BELOW JARS		57.33		Surface Losses		88	19 Tight hole / Fishing	9.50		
SPP (Calculated)		1270	86	HOURS: JAR MTR		134.5	CHEMICAL USAGE				21 Completion			
PUMP DATA				DRAG: UP DOWN		4	1	AMC Biocide G		2	21.1 Well Control			
#1: NAT 8-P-80		STROKE	9.0"	TORQUE: ON OFF		4900	1500	AMCPacLV		18	21.2 Other	0.75	40.00	
RATE		78		SURVEYS: MD INC' AZ'				Caustic Soda		2	TOTALS	24.00	440.50	
LINER		6.0"		795 1 0				Potassium Chloride		63	NON PRODUCTIVE		576.50	
#2: NAT 8-P-80		STROKE	9.0"	998 0.3 0				Soda Ash		3	PRODUCTS: USED	REC	ON SITE	
RATE		78						Sodium Sulphite		14	DIESEL (L)	8150	30000	64350
LINER		6.0"						Xan-Bore		7	BARITE (T)			26.0
#3: IDECO T-800		STROKE	8.5"								DRILL WATER	6000		28000
RATE		78									DAILY MUD COSTS			\$4,005.85
LINER		6.0"									CUM. MUD COSTS			\$43,295.65

HOURLY OPERATIONS SUMMARY 0000 to 2400

From	To	[IADC Code] Description
0:00	4:00	[2] Drilled 12 1/4" hole from 955m to 971m. 45klbs WOB, 80 RPM, 760GPM at 1940 psi., 6500 ft/lbs Torque. Av ROP: 4 m/hr.
4:00	4:45	[21.2] Investigate 300psi standpipe pressure loss. Diagnose fault in #1 mud pump.
4:45	9:45	[2] Drilled 12 1/4" hole from 971m to 981m. 45klbs WOB, 85 RPM, 700GPM at 1740 psi., 7000 ft/lbs Torque. Av ROP: 2.5 m/hr.
9:45	10:15	[7] Serviced rig.
10:15	18:00	[2] Drilled 12 1/4" hole from 971m to 997m. 45klbs WOB, 85 RPM, 725GPM at 1900 psi., 6000 ft/lbs Torque. Av ROP: 3.4 m/hr.
18:00	22:30	[2] Drilled 12 1/4" hole from 997m to 1011m. 45klbs WOB, 85 RPM, 725GPM at 1900 psi., 6000 ft/lbs Torque. Av ROP: 3.1m/hr.
22:30	23:00	[10] Ran wireline survey at 998m. 0.25 degree
23:00	0:00	[2] Drilled 12 1/4" hole from 1011m to 1014m. 45klbs WOB, 85 RPM, 725GPM at 1900 psi., 5000 ft/lbs Torque. Av ROP: 3 m/hr.

HOURLY OPERATIONS SUMMARY 0000 to 0600 on 09/09/18

0:00	6:00	Drilled 12 1/4" hole from 1014m to m. 45klbs WOB, 85 RPM, 730GPM at 1960 psi., 5000 ft/lbs Torque. Av ROP: 3.2m/hr.
------	------	---

POB-38: Ensign - 19, CTP - 4, OICS - 3, Halliburton - 4, OCS - 2, Site Med - 1, RMN - 1, WFT - 4

17,892 Hrs

RECEIVED:

DISPATCHED:

MAXIMUM GAS:

BACKGROUND GAS:

CONNECTION GAS:

TRIP GAS:

SUPERVISOR: Andy Phillis / Dave York

GEOLOGIST: Phil Allen / Peter Morris

RIG MGR: Simon Luke Bryant / Blake Smith



DAILY DRILLING REPORT

REPORT # 0209/09/2018

SPUD: 21/08/2018DAY # 066

WELL

Palm Valley 13

24:00 MD/ TVD

1092m BRT

DRILLED

78m

AFE DAYS / COST

46

\$13,467,756

RIG

ENSIGN 932

FORMATION

Carmichael Sst.

DAILY COST

\$137,202

CUM.

\$7,227,425

DAILY OP'S SUMMARY

Drilled 12 1/4" hole from 1014m to 1034m. Circulated bottoms up and tripped out of the hole for a bit change, due to downhole torque. Tripped in the hole with Bit #9. Drilled 12 1/4" hole from 1034m to 1092m.

FORECAST OP'S

Circulate hole clean. Trip out to run 9 5/8 casing. Rig up for and run 9 5/8" casing.

LAST CASING

340mm

13.375"

SET AT

249.0m

MAASP

BOP TEST

NIL

DUE

LAST L.T.I. 449 Days

SAFETY

1. PTSM: Current ops. Monitor mud pumps. Weekly inspections.

2. PTSM: Weekly Safety Meeting. Snake bites. Tripping ops.

WEATHER: Day

Fine

Night

Fine

BIT INFORMATION

WOB(Klb)

20

JET V(fps)

224

RPM

110

H S I

1.54

BIT NUMBER

9

8

Size (inches)

12.25

12.25

Make

Ulterra

HTC

Type

7 Blade PDC

KMX524T

IADC Code

Serial Number

40322

5281617

T.F.A. (in)

1.052

1.230

Depth In (m)

1034

758

Depth Out (m)

IN

1034

Total Meters

58

276

Hours IADC | OnBtm

10.5 | 9.4

46.5 | 42.6

ROP IADC | OnBtm

5.5 | 6.2

5.9 | 6.5

Condition Out

2 7 BT G E 0 BT TQ

FLOW DATA

RATE (gpm/lps)

736

46.4

AV - DP (fpm/mpm)

135

41

AV - DC (fpm/mpm)

223

68

SPP (psi/atm)

2100

143

SPP (Calculated)

1530

104

PUMP DATA

#1: NAT 8-P-80

STROKE

9.0"

RATE

78

LINER

6.0"

#2: NAT 8-P-80

STROKE

9.0"

RATE

78

LINER

6.0"

#3: IDECO T-800

STROKE

8.5"

RATE

78

LINER

6.0"

BHA # 5

TOOL

LENGTH

Bit - PDC

0.25

Near Bit Stabiliser

1.79

Float Sub

0.56

2x Drill Collar- 8"

18.62

Stabiliser

1.34

2x Drill Collar- 8"

18.81

Crossover Sub

0.79

16x Drill Collar- 6 1/2"

149.89

Crossover Sub

0.80

Drilling Jars

9.60

Crossover Sub

0.58

Drill Collar- 6 1/2"

9.39

Crossover Sub

0.80

8x Heavy Weight DP

75.15

BHA LENGTH

288.37

BHA WEIGHT

67.10

STRING WT

102.83

HOOK LOAD

124.00

WT BELOW JARS

57.33

HOURS: JAR | MTR

151.0

DRAG: UP | DOWN

7

1

TORQUE: ON | OFF

7800

1250

SURVEYS: MD | INC | AZ

998 | 0.3 | 0

MUD PROPERTIES

Mud Type

KCI Gel Polymer

Depth (m)

1034

1076

Density (ppg)

9.30

9.35

ECD (ppg)

9.40

9.45

Temp (° C)

47

50

Viscosity (sec)

38

38

PV / YP (cp/lb)

9 / 19

8 / 22

Gels (10s/m)

6 / 8

6 / 9

API Filt. (cc)

6.6

6.8

Sand (% Vol)

tr

tr

KCI (%)

3.2

3.5

pH (strip)

9.5

9.5

LGS %/Vol

5.8

6.0

Chlorides (ppm)

15750

16000

Surface Vol. (Bbls)

1247

Hole Vol.

485

Mud Mixed

217

New Hole Drilled

37.3

Downhole Losses

Surface Losses

166

CHEMICAL USAGE

AMCPacLV

9

Potassium Chloride

52

Soda Ash

3

Sodium Sulphite

4

Xan-Bore

6

OPERATION

HRS

CUM

1 Move/ Rig Up/Down

1119.50

2 Drilling

16.75

194.75

3 Wash / Ream

0.50

13.25

4 Coring

5 Circ & Condition

0.75

11.25

6 Tripping- Bit / Casing

4.75

28.50

6.1 Tripping- Other

19.50

6.2 Wiper Trip

6.3 Handle BHA

0.75

37.75

7 Service / Slip Line

0.50

8.25

8 Repairs

62.00

10 Survey

2.75

11 Logging

12 RU for Csg / Cmt

11.25

12.1 Run Casing

7.00

12.2 Cementing

9.25

13 Wait on Cement

9.50

14 BOP's / Wellhead

15 Test BOP / Formation

19 Tight hole / Fishing

9.50

21 Completion

21.1 Well Control

21.2 Other

40.00

TOTALS

24.00

464.50

NON PRODUCTIVE

576.50

PRODUCTS: USED | REC | ON SITE

DIESEL (L)

6150

58200

BARITE (T)

26.0

DRILL WATER

-24000

52000

DAILY MUD COSTS

\$2,440.37

CUM. MUD COSTS

\$45,557.40

HOURLY OPERATIONS SUMMARY 0000 to 2400

From

To

[IADC Code] Description

0:00

6:15

[2] Drilled 12 1/4" hole from 1014m to 1034m. 45klbs WOB, 85 RPM, 730GPM at 1960 psi., 5000 ft/lbs Torque. Av ROP: 3.2m/hr.

6:15

7:00

[5] Circulated bottoms up for bit trip.

7:00

9:15

[6] Tripped out of hole for bit trip (35klbs overpull at 579m).

9:15

10:00

[6.3] Broke out Bit #8. Made up Bit #9. Gauged stabilisers.

10:00

10:30

[7] Serviced rig.

10:30

13:00

[6] Tripped in hole to 972m.

13:00

13:30

[3] Washed down from 972m to 1034m.

13:30

18:00

[2] Drilled 12 1/4" hole from 1034m to 1062m. 20klbs WOB, 110 RPM, 730GPM at 2100 psi., 7500 ft/lbs Torque. Av ROP: 6.2m/hr.

18:00

0:00

[2] Drilled 12 1/4" hole from 1062m to 1092m. 20klbs WOB, 110 RPM, 730GPM at 2100 psi., 7500 ft/lbs Torque. Av ROP: 5 m/hr.

HOURLY OPERATIONS SUMMARY 0000 to 0600 on 10/09/18

0:00

6:00

Drilled 12 1/4" hole from 1092m to 1117m. 20klbs WOB, 110 RPM, 730GPM at 2100 psi., 5000 ft/lbs Torque. Av ROP: 4.1m/hr.

POB-42: Ensign - 19, CTP - 4, OICS - 3, Halliburton - 8, OCS - 2, Site Med - 1, RMN - 1, WFT - 4

18,396 Hrs

RECEIVED: 1 X 6 3/4" motor

DISPATCHED:

MAXIMUM GAS:

BACKGROUND GAS:

CONNECTION GAS:

TRIP GAS:

SUPERVISOR: Andy Phillis / Dave York

GEOLOGIST: Phil Allen / Peter Morris

RIG MGR: Simon Luke Bryant / Blake Smith

			DAILY DRILLING REPORT			REPORT # 023		12/09/2018	
						SPUD: 21/08/2018		DAY # 069	
WELL	Palm Valley 13	24:00 MD/ TVD	1119m BRT	DRILLED		AFE DAYS / COST	46	\$13,467,756	
RIG	ENSIGN 932	FORMATION	Stokes		DAILY COST	\$108,497	CUM.	\$7,659,280	
DAILY OP'S SUMMARY		Nipped up BOPE. Pressure tested BOPE. Ran in the hole with excess drill collars and laid out same.							
FORECAST OP'S		Tag top of cement. Drill out shoe track. Conduct FIT. Drill 8 1/2" directional hole.							
LAST CASING	244mm	9.625"	SET AT	1116.5m		MAASP		BOP TEST	NIL
LAST L.T.I.	452 Days		SAFETY	1. PTSM: Nipple up BOP. Pressure test BOP. Permit to work. Review JSA's. 2. TSM: Pressure testing. Trapped pressure. Look for leaks.			WEATHER: Day	Fine	
							Night	Fine	

BIT INFORMATION			BHA # 6		MUD PROPERTIES			OPERATION	HRS	CUM
WOB(Klb)	JET V(fps)		TOOL	LENGTH	Mud Type	KCl Gel Polymer				
RPM	H S I		Bit - PDC	0.24	Depth (m)	1119	1119			
BIT NUMBER	9	8	NOV Fluid Hammer- 6 3/4"	8.83	Density (ppg)	9.40	8.90			
Size (inches)	12.25	12.25	Stabiliser	1.69	ECD (ppg)					
Make	Ulterra	HTC	Float Sub	0.79	Temp (° C)					
Type	7 Blade PDC	KMX524T	Non Magnetic DC	9.16	Viscosity (sec)	38	38			
IADC Code			Directional Tools	1.93	PV / YP (cp/lb)	8 / 22	8 / 20			
Serial Number	40322	5281617	Crossover Sub	0.59	Gels (10s/m)	6 / 8	5 / 7			
T.F.A. (in)	1.052	1.230	Drill Collar- 6 1/2"	8.41	API Filt. (cc)	6.4	6.0			
Depth In (m)	1034	758	Crossover Sub	0.80	Sand (% Vol)	tr				
Depth Out (m)	1119	1034	18x Heavy Weight DP	169.32	KCl (%)	35.0	5.0			
Total Meters	85	276	Crossover Sub	0.45	pH (strip)	9.5	9.5			
Hours IADC OnBtm	17.3 14.9	46.5 42.6	Drilling Jars	8.98	LGS %/Vol	6.3	2.1			
ROP IADC OnBtm	4.9 5.7	5.9 6.5	Crossover Sub	0.30	Chlorides (ppm)	16000	21750			
Condition Out	2 1 CT N X 0 NO TD		6x Heavy Weight DP	56.42	Surface Vol. (Bbls)	768				
FLOW DATA			BHA LENGTH	267.91	Hole Vol.	256				
RATE (gpm/lps)			BHA WEIGHT	24.84	Mud Mixed					
AV - DP (fpm/mpm)			STRING WT	62.66	New Hole Drilled					
AV - DC (fpm/mpm)			HOOK LOAD		Downhole Losses					
SPP (psi/atm)			WT BELOW JARS	21.54	Surface Losses	156				
SPP (Calculated)			HOURS: JAR MTR	158.0	CHEMICAL USAGE					
PUMP DATA			DRAG: UP DOWN							
#1: NAT 8-P-80	STROKE	9.0"	TORQUE: ON OFF							
RATE			SURVEYS: MD INC' AZ'							
LINER	6.0"		1114 0.8 0							
#2: NAT 8-P-80	STROKE	9.0"								
RATE										
LINER	6.0"									
#3: IDECO T-800	STROKE	8.5"								
RATE										
LINER	6.0"									
HOURLY OPERATIONS SUMMARY 0000 to 2400										
From	To	[IADC Code] Description								
0:00	5:00	[14] Nipped up BOPE. Function tested same.								
5:00	7:30	[15] Rigged up for pressure testing BOP. Filled BOP, all lines. Made up drill pipe, X/O sub and cup tester.								
7:30	8:30	[15] Test #1. Wellhead connections, Annular, Mud pump valves, Outer blooie line valve, Choke manifold valves 1 & 2.								
8:30	9:30	[15] Laid out cup tester and made up test plug, landed out and flushed lines.								
9:30	13:30	[15] Tests #2 to #7. Mud pump isolation valves, Choke manifold valves, Kill line valves, Pipe rams, Kelly cocks and choke line valves.								
13:30	14:00	[15] Broke out and laid out dill pipe and hooked up test pump to choke manifold.								
14:00	15:00	[15] Test #8. Blind rams and check valve. (Check valve failed and was retested OK)								
15:00	16:30	[14] Installed clamp and hydraulic ram to RCD.								
16:30	16:45	[15] Attempted test on RCD flange - leaking.								
16:45	17:15	[21.2] Conducted muster drill. Tested ESD system - all engines shut down in less than 20seconds.								
17:15	18:00	[14] Re-tightened RCD flange bolts.								
18:00	18:15	[15] Tested RCD flange OK								
18:15	19:00	[14] Made up combo tool and landed out wear bushing.								
19:00	20:15	[15] Conducted Koomey drawdown test.								
20:15	21:15	[14] Installed flowline. Laid out pressure testing subs. Installed RCD bushing.								
21:15	0:00	[6.3] Tripped in with surplus drill collars and laid out same.								
HOURLY OPERATIONS SUMMARY 0000 to 0600 on 13/09/18										
0:00	2:30	Made up directional 8 1/2"BHA. Shallow tested motor. Scribed MWD.								
2:30	5:00	Ran in the hole to 287m picking up 24 singles of Hevi Wate drill pipe and drilling jars with crossover subs..								
5:00	6:00	Tripped in the hole to 693m MD.								
POB-44: Ensign - 20, CTP - 4, OICS - 3, Halliburton - 8, OCS - 1, Site Med - 2, RMN - 2, WFT - 4										19,956 Hrs
RECEIVED:					DISPATCHED:					
MAXIMUM GAS:			BACKGROUND GAS:			CONNECTION GAS:			TRIP GAS:	
SUPERVISOR: Andy Phillis / Dave York			GEOLOGIST: Phil Allen / Peter Morris			RIG MGR: Simon Luke Bryant / Blake Smith				

POB-43: Ensign - 20, CTP - 4, OICS - 3, Halliburton - 8, OCS - 1, Site Med - 1, RMN - 1, WFT - 5				20,472 Hrs
RECEIVED: Drill water		DISPATCHED:		
MAXIMUM GAS:		BACKGROUND GAS:		CONNECTION GAS:
TRIP GAS:				
SUPERVISOR: Andy Phillis / Mark Booth		GEOLOGIST: Phil Allen / Peter Morris		RIG MGR: Gary Hender / Blake Smith

POB-43: Ensign - 20, CTP - 4, OICS - 3, Halliburton - 8, OCS - 1, Site Med - 1, RMN - 1, WFT - 5				20,988 Hrs
RECEIVED: Drill Water			DISPATCHED:	
MAXIMUM GAS:		BACKGROUND GAS:		CONNECTION GAS:
TRIP GAS:				
SUPERVISOR: Andy Phillis / Mark Booth		GEOLOGIST: Phil Allen / Peter Morris		RIG MGR: Gary Hender / Blake Smith

POB-46: Ensign - 20, CTP - 4, OICS - 3, Halliburton - 8, OCS - 1, Site Med - 1, RMN - 1, WFT - 8				21,540 Hrs
RECEIVED: 7" Casing, Potable & Drill Water		DISPATCHED:		
MAXIMUM GAS:	0 U @ m	BACKGROUND GAS:	0 U	CONNECTION GAS:
0 U		0 U		TRIP GAS:
SUPERVISOR: Andy Phillis / Mark Booth		GEOLOGIST: Phil Allen / Peter Morris		RIG MGR: Gary Hender / Blake Smith

POB-46: Ensign - 20, CTP - 4, OICS - 3, Halliburton - 8, OCS - 1, Site Med - 1, RMN - 1, WFT - 8				22,092 Hrs
RECEIVED: Drill bit, fishing equipment and cement head.		DISPATCHED:		
MAXIMUM GAS:	0 U @ m	BACKGROUND GAS:	0 U	CONNECTION GAS:
0 U		0 U		TRIP GAS:
SUPERVISOR: Andy Phillis / Mark Booth		GEOLOGIST: Phil Allen / Peter Morris		RIG MGR: Gary Hender / Blake Smith

POB-48: Ensign - 20, CTP - 4, OICS - 3, Halliburton - 10, OCS - 1, Site Med - 1, RMN - 1, WFT - 8				23,232 Hrs
RECEIVED: Drill water		DISPATCHED:		
MAXIMUM GAS: 0.15 U @ 1470m		BACKGROUND GAS: 0.2 U		CONNECTION GAS: 0 U
TRIP GAS:				
SUPERVISOR: Andy Phillis / Mark Booth		GEOLOGIST: Phil Allen / Peter Morris		RIG MGR: Gary Hender / Blake Smith

POB-47: Ensign - 20, CTP - 4, OICS - 3, Halliburton - 9, OCS - 1, Site Med - 1, RMN - 1, WFT - 8				23,796 Hrs
RECEIVED: Drill bits			DISPATCHED:	
MAXIMUM GAS: 0.04 U @ 1514m		BACKGROUND GAS: 0.02 U		CONNECTION GAS: 0 U
TRIP GAS:				
SUPERVISOR: Kev Dau / Mark Booth		GEOLOGIST: Phil Allen / Peter Morris		RIG MGR: Gary Hender / Blake Smith

HOURLY OPERATIONS SUMMARY 0000 to 0600 on 21/09/18				
0:00	6:00	Directional drill 8 1/2" hole from 1675m to 1701m, WOB 11 Klbs, TQ 3000 ft/lbs @ 30 surface rpm, Air 2900scfm @ 440psi, 16 GPM mist rate, ROP 4.33 m/hr with connections.		
POB-48: Ensign - 20, CTP - 4, OICS - 3, Halliburton - 10, OCS - 1, Site Med - 1, RMN - 1, WFT - 8		24,372 Hrs		
RECEIVED: 7" Casing & mud Chemicals		DISPATCHED:		
MAXIMUM GAS: 0.05 U @ 1663m		BACKGROUND GAS: 0.03 U	CONNECTION GAS: 0 U	TRIP GAS:
SUPERVISOR: Kev Dau / Mark Booth		GEOLOGIST: Phil Allen / Peter Morris		RIG MGR: Gary Hender / Blake Smith

POB-50: Ensign - 20, CTP - 4, OICS - 3, Halliburton - 10, OCS - 1, Site Med - 1, RMN - 1, WFT - 8, Kenitec - 2				24,972 Hrs
RECEIVED: Mud chemicals		DISPATCHED: Drilling equipment to BECC's.		
MAXIMUM GAS: 0.06 U @ 1728m		BACKGROUND GAS: 0.02 U		CONNECTION GAS: 0 U
TRIP GAS:				
SUPERVISOR: Kev Dau / Mark Booth		GEOLOGIST: Phil Allen / Peter Morris		RIG MGR: Gary Hender / Blake Smith



DAILY DRILLING REPORT

REPORT # 03322/09/2018

SPUD: 21/08/2018DAY # 079

WELL

Palm Valley 13

24:00 MD/ TVD

1753 / 1707m BRT

DRILLED

20m

AFE DAYS / COST

46

\$13,467,756

RIG

ENSIGN 932

FORMATION

Lower Stairway Sandstone.

DAILY COST

\$170,001

CUM.

\$9,258,321

DAILY OP'S SUMMARY

Pulled out of hole, laid out Bit, NMDC and DHM, replaced service winch line, made up BHA and ran in hole unloading well, circulated and washed to bottom, broke in bit and drilled ahead 8 1/2" directional hole 1753mMD

FORECAST OP'S

Drill ahead 8 1/2" directional hole.

LAST CASING

244mm

9.625"

SET AT

1116.5m

FIT

14.2 EMW

MAASP

924psi

BOP TEST

12/09

DUE

26/09

LAST L.T.I.

462 Days

SAFETY

1. Discussed incident with service winch cable, tripping and UBD drilling operations.

2. Discussed incident with service winch cable, unloading well and UBD drilling operations.

WEATHER: Day

Fine

Night

Fine

BIT INFORMATION			BHA # 15		MUD PROPERTIES		OPERATION	HRS	CUM
WOB(Klb)	10-15	JET V(fps)	TOOL	LENGTH	Mud Type	Stored mud			
RPM	30	H S I	Bit - PDC	0.29	Depth (m)				
BIT NUMBER	13	12	Down Hole Motor	7.76	Density (ppg)	9.35	1 Move/ Rig Up/Down		1122.75
Size (inches)	8.5	8.5	Float Sub	0.68	ECD (ppg)		2 Drilling	3.00	358.00
Make	Ulterra	HTC	Stabiliser	1.69	Temp (° C)		3 Wash / Ream	1.25	17.00
Type	CF713	KM526	Non Magnetic DC	7.03	Viscosity (sec)	42	4 Coring		
IADC Code			Directional Tools	1.95	PV / YP (cp/lb)	10 / 21	5 Circ & Condition	6.50	31.75
Serial Number	39970	5285339	Crossover Sub	0.96	Gels (10s/m)	7 / 9	6 Tripping- Bit / Casing	5.75	63.75
T.F.A. (in)	0.773	0.632	3x Drill Collar- 6 1/2"	28.33	API Filt. (cc)	6.0	6.1 Tripping- Other		25.75
Depth In (m)	1733	1442	Crossover Sub	0.80	Sand (% Vol)	0.1	6.2 Wiper Trip		
Depth Out (m)	IN	1733	18x Heavy Weight DP	169.32	KCl (%)	4.5	6.3 Handle BHA	5.50	53.75
Total Meters	20	291	Crossover Sub	0.45	pH (strip)	9.5	7 Service / Slip Line		11.75
Hours IADC OnBtm	3 2.1	77.5 63.6	Drilling Jars	8.98	LGS %/Vol	5.6	8 Repairs	2.00	66.25
ROP IADC OnBtm	6.7 9.5	3.8 4.6	Crossover Sub	0.30	Chlorides (ppm)	20000	10 Survey		2.75
Condition Out	3 3 BT S E 2 WT PR		6x Heavy Weight DP	56.42	Surface Vol. (Bbls)	1555	11 Logging		
FLOW DATA			BHA LENGTH	284.96	Hole Vol.	388	12 RU for Csg / Cmt		16.50
RATE (gpm/lps)			BHA WEIGHT	29.56	Mud Mixed	219	12.1 Run Casing		16.50
AV - DP (fpm/mpm)			STRING WT	94.84	New Hole Drilled	4.6	12.2 Cementing		12.50
AV - DC (fpm/mpm)			HOOK LOAD	145.00	Downhole Losses		13 Wait on Cement		16.00
SPP (psi/atm)	580	39	WT BELOW JARS	26.26	Surface Losses	-5	14 BOP's / Wellhead		18.00
SPP (Calculated)			HOURS: JAR MTR	156.5 77.5	CHEMICAL USAGE		15 Test BOP / Formation		13.00
PUMP DATA			DRAG: UP DOWN	12 5	AMC Biocide G	1	19 Tight hole / Fishing		9.50
#1: NAT 8-P-80	STROKE	9.0"	TORQUE: ON OFF	2000 4000	AMC Defoamer S	1	21 Completion		
RATE			SURVEYS: MD INC' AZ'		AMCPacLV	1	21.1 Well Control		
LINER	6.0"		1712.6 31.5 39.4		Soda Ash	1	21.2 Other		40.50
#2: NAT 8-P-80	STROKE	9.0"	1750.5 34.5 33.9		AMC Superfoam	19	TOTALS	24.00	773.25
RATE							NON PRODUCTIVE	2.00	588.50
LINER	6.0"						PRODUCTS: USED REC ON SITE		
#3: IDECO T-800	STROKE	8.5"					DIESEL (L)	2400	58900
RATE							BARITE (T)		26.0
LINER	6.0"						DRILL WATER		120000
DAILY MUD COSTS									\$1,319.19
CUM. MUD COSTS									\$65,780.89

HOURLY OPERATIONS SUMMARY 0000 to 2400

From	To	[IADC Code] Description
0:00	2:00	[6] PJSM, Pulled out of hole from 595mMD to surface.
2:00	5:00	[6.3] Inspected and laid out BHA, Bit 3/3/BT/UG, DHM 6mm end float and lateral movement, index sub washed, dart float damaged.
5:00	7:00	[5] PJSM, prepared for mud drilling, Installed flowline and bell nipple.
7:00	9:00	[8] NPT- PJSM, write and review JSA, install replacement service winch cable in mast.
9:00	11:00	[5] DHM and mist pump on location prepared for air drilling, removed flowline and installed blanking flange.
11:00	13:30	[6.3] Picked up and made up Directional BHA and ran in hole to 20mMD
13:30	15:00	[6] Ran in hole from 20m to 568mMD.
15:00	15:45	[5] PJSM, installed RCD.
15:45	16:00	[5] Unloaded well @ 568mMD injection pressure 1306psi 2900 CFM.
16:00	17:00	[6] Ran in hole from 568m to 1144mMD.
17:00	17:45	[5] Unloaded well @ 1144mMD injection pressure 898psi 2900 CFM.
17:45	19:00	[6] Ran in hole from 1144m to 1662mMD.
19:00	19:45	[5] Unloaded well @ 1662mMD injection pressure 1118psi 2900 CFM.
19:45	21:00	[3] Washed ream from 1662mMD to 1733mMD.
21:00	0:00	[2] Directional drill 8 1/2" hole from 1733m to 1753m, WOB 11 Klbs, TQ 3500 ft/lbs @ 30 surface rpm, Air 2900 scfm @ 585 psi, 20 GPM mist rate, ROP 6 m/hr.

HOURLY OPERATIONS SUMMARY 0000 to 0600 on 23/09/18

0:00	6:00	Directional drill 8 1/2" hole from 1753m to 1790m, WOB 11 Klbs, TQ 3500 ft/lbs @ 30 surface rpm, Air 2960 scfm @ 755 psi, 22 GPM mist rate, ROP 6.1 m/hr.
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POB-54: Ensign - 20, CTP - 4, OICS - 3, Halliburton - 12, OCS - 1, Site Med - 1, RMN - 1, WFT - 7, Kenitec - 5

25,620 Hrs

RECEIVED: Mist pump

DISPATCHED:

MAXIMUM GAS: 2 U @ 1739m

BACKGROUND GAS: .04 U

CONNECTION GAS: 11.5 U

TRIP GAS:

SUPERVISOR: Kev Dau / Mark Booth

GEOLOGIST: Phil Allen / Peter Morris

RIG MGR: Gary Hender / Blake Smith

			DAILY DRILLING REPORT				REPORT # 034		23/09/2018			
							SPUD: 21/08/2018		DAY # 080			
WELL	Palm Valley 13	24:00 MD/ TVD	1845 / 1782m BRT	DRILLED	92m	AFE DAYS / COST	46	\$13,467,756				
RIG	ENSIGN 932	FORMATION	Horn Valley Siltstone		DAILY COST	\$124,761	CUM.	\$9,395,083				
DAILY OP'S SUMMARY		Drilled ahead 8 1/2" directional hole from 1753mMD to 1845mMD, circulated, pulled out of hole to 1101mMD, preformed gas flow test, ran in hole.										
FORECAST OP'S		Displace well to drill mud, install flow line, wash to 1845mMD, circulate hole clean, pull out of hole and lay down directional BHA, make up and run casing scraper.										
LAST CASING	244mm	9.625"	SET AT	1116.5m	FIT	14.2 EMW	MAASP	924psi	BOP TEST	12/09	DUE	26/09
LAST L.T.I. 463 Days		SAFETY		1. Discussed connection gas and air drilling, team work and housekeeping. 2. Discussed checking equipment, heat stress, vehicle speeds around camp & lease, bullying and harassment policies.				WEATHER: Day		Fine		
								Night		Fine		

BIT INFORMATION				BHA # 15		MUD PROPERTIES		OPERATION		HRS	CUM
WOB(Klb)	7-9	JET V(fps)		TOOL	LENGTH	Mud Type	KCl Gel Polymer	1 Move/ Rig Up/Down			1122.75
RPM	30	H S I		Bit - PDC	0.29	Depth (m)	1845	2 Drilling	13.75		371.75
BIT NUMBER	13	12		Down Hole Motor	7.76	Density (ppg)	9.35	3 Wash / Ream			17.00
Size (inches)	8.5	8.5		Float Sub	0.68	ECD (ppg)		4 Coring			
Make	Ulterra	HTC		Stabiliser	1.69	Temp (° C)		5 Circ & Condition	2.00		33.75
Type	CF713	KM526		Non Magnetic DC	7.03	Viscosity (sec)	40	6 Tripping- Bit / Casing			63.75
IADC Code				Directional Tools	1.95	PV / YP (cp/lb)	10 / 20	6.1 Tripping- Other	1.75		27.50
Serial Number	39970	5285339		Crossover Sub	0.96	Gels (10s/m)	5 / 7	6.2 Wiper Trip	1.50		1.50
T.F.A. (in)	0.773	0.632		3x Drill Collar- 6 1/2"	28.33	API Filt. (cc)	6.2	6.3 Handle BHA			53.75
Depth In (m)	1733	1442		Crossover Sub	0.80	Sand (% Vol)	0.1	7 Service / Slip Line	0.50		12.25
Depth Out (m)	1845	1733		18x Heavy Weight DP	169.32	KCl (%)	4.5	8 Repairs			66.25
Total Meters	112	291		Crossover Sub	0.45	pH (strip)	9.0	10 Survey			2.75
Hours IADC OnBtm	16.8 11.5	77.5 63.6		Drilling Jars	8.98	LGS %/Vol	5.5	11 Logging			
ROP IADC OnBtm	6.7 9.7	3.8 4.6		Crossover Sub	0.30	Chlorides (ppm)	19500	12 RU for Csg / Cmt			16.50
Condition Out				6x Heavy Weight DP	56.42	Surface Vol. (Bbls)	1555	12.1 Run Casing			16.50
FLOW DATA				BHA LENGTH	284.96	Hole Vol.	408	12.2 Cementing			12.50
RATE (gpm/lps)				BHA WEIGHT	29.56	Mud Mixed	577	13 Wait on Cement			16.00
AV - DP (fpm/mpm)				STRING WT	98.93	New Hole Drilled	21.2	14 BOP's / Wellhead			18.00
AV - DC (fpm/mpm)				HOOK LOAD	145.00	Downhole Losses		15 Test BOP / Formation	4.50		17.50
SPP (psi/atm)	790	54		WT BELOW JARS	26.26	Surface Losses	556	19 Tight hole / Fishing			9.50
SPP (Calculated)				HOURS: JAR MTR	170.5 77.5	CHEMICAL USAGE		21 Completion			
PUMP DATA				DRAG: UP DOWN	12 5	AMC Defoamer S	30	21.1 Well Control			
#1: NAT 8-P-80	STROKE	9.0"		TORQUE: ON OFF	3000 3500	AMCPacLV	10	21.2 Other			40.50
RATE				SURVEYS: MD INC' AZ'		Caustic Soda	1	TOTALS	24.00	797.25	
LINER	6.0"			1750.5 34.5 33.9		Soda Ash	5	NON PRODUCTIVE		588.50	
#2: NAT 8-P-80	STROKE	9.0"		1769.7 35.3 29.9		Xan-Bore	20	PRODUCTS: USED REC ON SITE			
RATE				1789 35.6 25.5		Hydro 327	7	DIESEL (L)	9500	10000	59400
LINER	6.0"			1808.4 36.1 21.5		AMC Superfoam	27	BARITE (T)			26.0
#3: IDECO T-800	STROKE	8.5"		1827.7 36.2 15.4				DRILL WATER	10000		110000
RATE								DAILY MUD COSTS			\$7,079.47
LINER	6.0"							CUM. MUD COSTS			\$72,860.36

HOURLY OPERATIONS SUMMARY 0000 to 2400		
From	To	[IADC Code] Description
0:00	6:00	[2] Directional drill 8 1/2" hole from 1753m to 1790m, WOB 11 Klbs, TQ 3500 ft/lbs @ 30 surface rpm, Air 2960 scfm @ 755 psi, 22 GPM mist rate, ROP 6.1 m/hr.
6:00	10:15	[2] Directional drill 8 1/2" hole from 1790m 1820m, WOB 11 Klbs, TQ 3450 ft/lbs @ 30 surface rpm, Air 2960 scfm @ 755 psi, 22 GPM mist rate, ROP 7 m/hr with connections.
10:15	10:45	[7] PJSM, serviced rig.
10:45	11:15	[5] PJSM, changed out RCD.
11:15	14:45	[2] Directional drill 8 1/2" hole from 1820m 1845m, WOB 11 Klbs, TQ 3450 ft/lbs @ 30 surface rpm, Air 2960 scfm @ 755 psi, 22 GPM mist rate, ROP 7 m/hr with connections.
14:45	15:45	[5] Pumped 10 bbl sweeps and circulated well clean.
15:45	17:30	[6.1] Pulled out of hole from 1845mMD to 1101mMD.
17:30	18:00	[15] PJSM, flow testing gas in well.
18:00	19:00	[15] Shut in well and monitor pressure build up, Maximum pressure 4.3 psi.
19:00	20:00	[15] Flow tested well through blooie line flow meter fitted with 0.5 orifice plate, 4.3 - 1.2 psi, 0.037 - 0.017 MMCF flow rate.
20:00	22:00	[15] Shut in well and monitored pressure build up, Maximum pressure 7 psi. Bled off and open well.
22:00	23:30	[6.2] PJSM, ran in hole from 1101mMD to 1733mMD on primary jet.
23:30	0:00	[5] Pumped 30 bbl water spacer and drill mud.

HOURLY OPERATIONS SUMMARY 0000 to 0600 on 24/09/18		
0:00	1:00	Continued to displace well to drill mud, flow checked.
1:00	1:30	Circulated well through poor boy degasser to even mud weights. Flow check - OK.
1:30	2:45	PJSM, removed RCD blanking flange and installed flow line.
2:45	3:15	PJSM, ran in hole from 1733mMD to 1845mMD.
3:15	4:30	Circulated hole clean. Trip gas 1.3%. Flow checked - OK.
4:30	6:00	PJSM, released clamp and removed RCD, pulled out of hole to run casing.

POB-48: Ensign - 20, CTP - 4, OICS - 3, Halliburton - 11, OCS - 1, Site Med - 1, RMN - 1, WFT - 7, Kenitec - 0		26,196 Hrs
RECEIVED:		DISPATCHED:
MAXIMUM GAS:	1823 U @ 21.8m	BACKGROUND GAS: 0.04 U
SUPERVISOR: Kev Dau / Mark Booth		CONNECTION GAS: 11.5 U
GEOLOGIST: Phil Allen / Peter Morris		TRIP GAS:
RIG MGR: Gary Hender / Blake Smith		

		DAILY DRILLING REPORT				REPORT # 035		24/09/2018				
						SPUD: 21/08/2018		DAY # 081				
WELL	Palm Valley 13	24:00 MD/ TVD	1845 / 1782m BRT	DRILLED		AFE DAYS / COST	46	\$13,467,756				
RIG	ENSIGN 932	FORMATION	Horn Valley Siltstone		DAILY COST	\$126,943	CUM.	\$9,530,026				
DAILY OP'S SUMMARY		Displaced well to drill mud, installed flow line, washed to 1845mMD, circulated hole clean, pulled out of hole and laid down directional BHA, made up and ran casing scraper.										
FORECAST OP'S		Run CBL to 9 5/8" Casing shoe, slip and cut drilling line, rig up and run 7" liner, circulate and cement liner.										
LAST CASING	244mm	9.625"	SET AT	1116.5m	FIT	14.2 EMW	MAASP	933psi	BOP TEST	12/09	DUE	26/09
LAST L.T.I. 464 Days		SAFETY		1. Discussed displacing well to mud and hazzards, team work, installing flowline. 2. Discussed running casing scaper, drifting pipe, tripping operations and heat stress.				WEATHER: Day		Fine		
								Night		Fine		

BIT INFORMATION			BHA # 16		MUD PROPERTIES		OPERATION		HRS	CUM
WOB(Klb)	JET V(fps)	194	TOOL	LENGTH	Mud Type	KCl Gel Polymer	1 Move/ Rig Up/Down			1122.75
RPM	H S I	1.51	Bit - Tri Cone	0.24	Depth (m)	1845	2 Drilling			371.75
BIT NUMBER	13	12	Casing Scraper	1.22	Density (ppg)	9.30	3 Wash / Ream			17.00
Size (inches)	8.5	8.5	Bit Sub	0.91	ECD (ppg)		4 Coring			
Make	Ulterra	HTC	Crossover Sub	0.80	Temp (° C)	28	5 Circ & Condition	5.25		39.00
Type	CF713	KM526	24x Heavy Weight DP	225.74	Viscosity (sec)	34	6 Tripping- Bit / Casing	4.75		68.50
IADC Code					PV / YP (cp/lb)	5 / 10	6.1 Tripping- Other	9.75		37.25
Serial Number	39970	5285339			Gels (10s/m)	2 / 3	6.2 Wiper Trip	0.50		2.00
T.F.A. (in)	0.773	0.632			API Filt. (cc)	11.7	6.3 Handle BHA	3.75		57.50
Depth In (m)	1733	1442			Sand (% Vol)	0.1	7 Service / Slip Line			12.25
Depth Out (m)	1845	1733			KCl (%)	3.6	8 Repairs			66.25
Total Meters	112	291			pH (strip)	8.5	10 Survey			2.75
Hours IADC OnBtm	16.8 11.5	77.5 63.6			LGS %/Vol	5.7	11 Logging			
ROP IADC OnBtm	6.7 9.7	3.8 4.6			Chlorides (ppm)	17500	12 RU for Csg / Cmt			16.50
Condition Out	1 1 CT G X 0 WT TD				Surface Vol. (Bbls)	778	12.1 Run Casing			16.50
FLOW DATA			BHA LENGTH	228.91	Hole Vol.	445	12.2 Cementing			12.50
RATE (gpm/lps)	467	29.5	BHA WEIGHT	18.69	Mud Mixed		13 Wait on Cement			16.00
AV - DP (fpm/mpm)	159	48	STRING WT	11.04	New Hole Drilled		14 BOP's / Wellhead			18.00
AV - DC (fpm/mpm)	#DIV/0!	#DIV/0!	HOOK LOAD		Downhole Losses		15 Test BOP / Formation			17.50
SPP (psi/atm)	800	54	WT BELOW JARS		Surface Losses	777	19 Tight hole / Fishing			9.50
SPP (Calculated)			HOURS: JAR MTR	170.5 77.5	CHEMICAL USAGE		21 Completion			
PUMP DATA			DRAG: UP DOWN		AMC Biocide G	1	21.1 Well Control			
#1: NAT 8-P-80	STROKE	9.0"	TORQUE: ON OFF		Xan-Bore	2	21.2 Other			40.50
RATE			SURVEYS: MD INC' AZ'				TOTALS	24.00	821.25	
LINER	6.0"	DEPTH: 1845m	1827.7 36.2 15.4				NON PRODUCTIVE		588.50	
#2: NAT 8-P-80	STROKE	9.0"					PRODUCTS: USED REC ON SITE			
RATE	75	SCR: 240 @ 30 Stks					DIESEL (L)	5900		53500
LINER	6.0"	DEPTH: 1845m					BARITE (T)			26.0
#3: IDECO T-800	STROKE	8.5"					DRILL WATER	20000		90000
RATE	75	SCR: 240 @ 30 Stks					DAILY MUD COSTS			\$261.07
LINER	6.0"	DEPTH: 1845m					CUM. MUD COSTS			\$73,121.43

HOURLY OPERATIONS SUMMARY 0000 to 2400		
From	To	[IADC Code] Description
0:00	1:00	[5] Continued to displace well to drill mud, flow checked.
1:00	1:30	[5] Circulated well through poor boy degasser to even mud weights. Flow check - OK.
1:30	2:45	[5] PJSM, removed RCD blanking flange and installed flow line.
2:45	3:15	[6.2] PJSM, ran in hole from 1733mMD to 1845mMD.
3:15	4:30	[5] Circulated hole clean. Trip gas 1.3%. Flow checked - OK.
4:30	6:00	[6] PJSM, released clamp and removed RCD, pulled out of hole from 1845mMD to 1745mMD to run casing.
6:00	9:00	[6] Flow check, pumped slug and pulled out of hole from 1745mMD to 280mMD.
9:00	9:15	[6] Remove RCD bearing from stands and rack back in mast.
9:15	11:15	[6.3] Pulled out of hole from 280mMD to 22mMD, laid out jars and 6 1/2" drill collars.
11:15	13:00	[6.3] PJSM, laid out direction BHA, bit graded 1/1CT/G/XWT-DL/TD, DHM end float 3 mm.
13:00	16:45	[6.1] Made up casing scraper BHA and ran in hole from surface to 700mMD.
16:45	17:00	[6.1] Pulled out of hole from 700mMD to 600mMD, working casing scraper through liner hanger seating area.
17:00	18:00	[6.1] Ran in hole from 600mMD to 1085mMD.
18:00	19:15	[5] Circulated casing clean @ 600 gpm.
19:15	22:30	[6.1] Pulled out of hole from 1085MMD to 288mMD, drifting drill pipe.
22:30	0:00	[6.1] Pulled out of hole scraper run BHA from 288mMD to surface, laid out bit and casing scraper.

HOURLY OPERATIONS SUMMARY 0000 to 0600 on 25/09/18		
From	To	Description
0:00	1:00	Prepared floor for CBL, held PJSM with Kinetic operators, rigged up to run CBL.
1:00	4:00	Ran CBL log from 9 5/8" casing shoe 1116.5m to surface, laid out and rigged down Kinetic.
4:00	5:30	PJSM, slipped and cut drill line.
5:30	6:00	Serviced rig and replaced bolts on TDS torque tube.

POB-48: Ensign - 20, CTP - 4, OICS - 3, Halliburton - 11, OCS - 1, Site Med - 1, RMN - 1, WFT - 7, Kenitec - 0								26,772 Hrs			
RECEIVED:					DISPATCHED:						
MAXIMUM GAS:		0.6 U @ 1845m		BACKGROUND GAS:				CONNECTION GAS:			
SUPERVISOR: Kev Dau / Mark Booth					GEOLOGIST: Phil Allen / Peter Morris			RIG MGR: Gary Hender / Blake Smith			

POB-50: Ensign - 20, CTP - 4, OICS - 4, Halliburton - 11, OCS - 2, Site Med - 1, RMN - 1, WFT - 7, Kenitec - 0				27,372 Hrs	
RECEIVED:			DISPATCHED:		
MAXIMUM GAS: 0 U @ m		BACKGROUND GAS: 0 U		CONNECTION GAS: 0 U	
TRIP GAS:					
SUPERVISOR: Kev Dau / Mark Booth		GEOLOGIST: Phil Allen / Peter Morris		RIG MGR: Gary Hender / Blake Smith	

			DAILY DRILLING REPORT			REPORT # 037		26/09/2018	
						SPUD: 21/08/2018		DAY # 083	
WELL	Palm Valley 13	24:00 MD/ TVD	1845 / 1782m BRT	DRILLED		AFE DAYS / COST	46	\$13,467,756	
RIG	ENSIGN 932	FORMATION	Horn Valley Siltstone		DAILY COST	\$292,419	CUM.	\$10,292,116	
DAILY OP'S SUMMARY		Circulated and cemented casing as per program, released liner hanger and circulated out excess cement, pulled out of hole and laid out liner hanger tool, Pressure tested BOPE, rigged up to run casing.							
FORECAST OP'S		RIH with 7" casing Tie back assembly, space out, pressure test DDV lines, circulate casing to brine, land out tie back string, pressure test, make up drill out BHA and RIH.							
LAST CASING	178mm	7.0"	SET AT	1842.0m	MAASP		BOP TEST	26/09	DUE 10/10
LAST L.T.I. 466 Days		SAFETY		1. Discussed hazards with cementing, trapped pressure, housekeeping, fit for work. 2. Discussed recent incidents, heat stress, hydration, pressure testing BOPE.			WEATHER: Day Fine Night Fine		

BIT INFORMATION			TOOL		LENGTH		MUD PROPERTIES		OPERATION		HRS	CUM
WOB(Klb)		JET V(fps)					Mud Type	KCl Gel Polymer	1 Move/ Rig Up/Down			1122.75
RPM		H S I					Depth (m)	1845	2 Drilling			371.75
BIT NUMBER	13	12					Density (ppg)	9.25	3 Wash / Ream			17.00
Size (inches)	8.5	8.5					ECD (ppg)		4 Coring			
Make	Ulterra	HTC					Temp (° C)		5 Circ & Condition	0.75		40.75
Type	CF713	KM526					Viscosity (sec)	31	6 Tripping- Bit / Casing			68.50
IADC Code							PV / YP (cp/lb)	3 / 6	6.1 Tripping- Other	2.00		39.25
Serial Number	39970	5285339					Gels (10s/m)	1 / 2	6.2 Wiper Trip			2.00
T.F.A. (in)	0.773	0.632					API Filt. (cc)	17.0	6.3 Handle BHA			57.50
Depth In (m)	1733	1442					Sand (% Vol)	0.1	7 Service / Slip Line			13.75
Depth Out (m)	1845	1733					KCl (%)	3.5	8 Repairs	0.50		66.75
Total Meters	112	291					pH (strip)	8.5	10 Survey			2.75
Hours IADC OnBtm	16.8 11.5	77.5 63.6					LGS %/Vol	5.4	11 Logging			4.00
ROP IADC OnBtm	6.7 9.7	3.8 4.6					Chlorides (ppm)	18000	12 RU for Csg / Cmt	6.50		26.00
Condition Out	1 1 CT G X 0 WT TD						Surface Vol. (Bbls)	874	12.1 Run Casing			31.00
FLOW DATA			BHA LENGTH				Hole Vol.		449	12.2 Cementing	2.75	15.25
RATE (gpm/lps)			BHA WEIGHT				Mud Mixed		170	13 Wait on Cement		16.00
AV - DP (fpm/MPM)			STRING WT				New Hole Drilled			14 BOP's / Wellhead		18.00
AV - DC (fpm/MPM)			HOOK LOAD				Downhole Losses			15 Test BOP / Formation	11.50	29.00
SPP (psi/atm)			WT BELOW JARS				Surface Losses		404	19 Tight hole / Fishing		9.50
SPP (Calculated)			HOURS: JAR MTR		170.5	77.5	CHEMICAL USAGE			21 Completion		
PUMP DATA			DRAG: UP DOWN				AMC Biocide G		3	21.1 Well Control		
#1: NAT 8-P-80	STROKE	9.0"	TORQUE: ON OFF				Potassium Chloride		28	21.2 Other		40.50
RATE			SURVEYS: MD INC* AZ*				Hydro 327		1	TOTALS	24.00	869.25
LINER	6.0"		1827.7 36.2 15.4							NON PRODUCTIVE	0.50	635.25
#2: NAT 8-P-80	STROKE	9.0"								PRODUCTS: USED REC ON SITE		
RATE										DIESEL (L)	900	51000
LINER	6.0"									BARITE (T)		20.0
#3: IDECO T-800	STROKE	8.5"								DRILL WATER	40000	60000 100000
RATE										DAILY MUD COSTS \$771.03		
LINER	6.0"									CUM. MUD COSTS \$74,812.46		

HOURLY OPERATIONS SUMMARY 0000 to 2400			
From	To	IADC Code	Description
0:00	1:15	[12]	Circulated, Held PJSM with Halliburton, pumped tune spacer and installed cement head.
1:15	3:45	[12.2]	Mixed and pumped 92.8 bbls lead @ 13.5 ppg and 36 bbls Tail @ 15.8 ppg, displaced 167 bbls, bumped plug and tested casing 2000 psi, 1.2 bbls return, floats held.
3:45	4:00	[12.2]	Released liner hanger as instructed by Halliburton.
4:00	4:45	[5]	Circulated out cement, 30 bbls cement to surface.
4:45	5:30	[12]	PJSM, Layout cement head and rig down cementing lines.
5:30	7:30	[6.1]	POOH racked back D/P
7:30	8:30	[12]	PJSM Layout Halliburton Versa flex Liner Hanger running tool
8:30	9:00	[12]	Picked up Halliburton cementing head broke out X/O subs and layout
9:00	10:00	[15]	Made up combo tool and removed wear bushing, M/U jetting tool and flushed BOP's and well head area
10:00	12:00	[15]	Conducted pressure testing well control equipment Test #1 Well head, Annular, Choke valve # 1, # 2 and outer blooie line valve.
12:00	13:15	[15]	PJSM, Pressure tested #2 MP #3 4" isolation valve, Annular, Choke valves # 3,8,9.
13:15	14:00	[15]	Pressure tested #3, MP#2 Isolation, Pipe rams, Choke valves # 4,6,8.
14:00	15:00	[15]	Pressure tested #4, MP 4" Isolator, Pipe rams, Choke valves # 6,7,11.
15:00	15:30	[15]	Pressure tested #5, Kill line isolator, upper standpipe 4", Pipe rams, Choke valves # 6,10,12.
15:30	16:30	[15]	Pressure tested #6, Pipe rams, Outer kill, HCR, Upper Kelly cock.
16:30	17:00	[8]	NPT- Pressure tested #7, FOSV failed test.
17:00	19:00	[12]	Rig up Weatherford DDV hydraulic line sheave to monkey board and run control lines.
19:00	20:30	[15]	Pressure tested #8, MP#3 isolator, pipe rams, manual HCR, inner kill. Laid out test assembly, made up hanger to landing joint and prepared XT39 drill pipe.
20:30	23:00	[15]	Pressure tested #9, Blind rams, kill check valve, casing, choke 6,10,12. Rigged down testing equipment and lined up valves.
23:00	0:00	[12]	Rigged up to run 7" casing Tie back Assembly.

HOURLY OPERATIONS SUMMARY 0000 to 0600 on 27/09/18			
From	To	Description	
0:00	1:30	Continued to rig up for casing, PJSM, picked up landing joint and hanger, landed hanger, noted landing height, laid out hanger.	
1:30	2:00	Picked up seal assembly and ran in hole from surface to 44m.	
2:00	3:45	PJSM, picked up and ran DDV, installed control lines and pressure tested to 3500 psi for 10 min open and close lines, function tested DDV.	
3:45	6:00	Ran in hole with Tie Back assembly.	

POB-53: Ensign - 20, CTP - 4, OICS - 4, Halliburton - 11, OCS - 2, Site Med - 2, RMN - 2, WFT - 7, Cactus - 1		28,008 Hrs
RECEIVED: Potable water		DISPATCHED:
MAXIMUM GAS: 0 U @ m	BACKGROUND GAS: 0 U	CONNECTION GAS: 0 U TRIP GAS:
SUPERVISOR: Kev Dau / Mark Booth		GEOLOGIST: Phil Allen / Peter Morris RIG MGR: Gary Hender / Blake Smith

		DAILY DRILLING REPORT				REPORT # 038		27/09/2018				
						SPUD: 21/08/2018		DAY # 084				
WELL	Palm Valley 13	24:00 MD/ TVD	1845 / 1782m BRT	DRILLED		AFE DAYS / COST	46	\$13,467,756				
RIG	ENSIGN 932	FORMATION	Horn Valley Siltstone		DAILY COST	\$281,859	CUM.	\$10,573,975				
DAILY OP'S SUMMARY Ran in hole with 7" casing Tie back assembly, spaced out, pressure tested DDV lines, circulated casing to brine, landed out tie back string, pressure tested control lines and casing, made up drill out BHA and ran in hole.		FORECAST OP'S Run in hole with drill out BHA, drill out shoe track and 3 meters formation, circulate, leak off test, displace well to water, pull out of hole, run CBL over 7" casing liner, make up 6 1/8" Direction BHA and run in hole unloading well.										
LAST CASING		178mm	7.0"	SET AT	1842.0m		MAASP		BOP TEST	26/09	DUE	10/10
LAST L.T.I. 467 Days		SAFETY 1. Discussed DDV operations and hazards with tripping, snakes, buddy system for short change and working with third party. 2. Discussed winch line incident and tugger new SWL, update on operations, fit for work and housekeeping.		WEATHER: Day Fine Night Fine								

BIT INFORMATION				BHA # 17		MUD PROPERTIES		OPERATION		HRS	CUM			
WOB(Klb)		JET V(fps)		TOOL		LENGTH		Mud Type	KCl Gel Polymer	1 Move/ Rig Up/Down		1122.75		
RPM		H S I		Bit - Tri Cone		0.19		Depth (m)	1845	2 Drilling		371.75		
BIT NUMBER		13	12	Bit Sub		0.88		Density (ppg)	9.25	3 Wash / Ream		17.00		
Size (inches)		8.5	8.5	9x Drill Collar- 4 3/4"		83.69		ECD (ppg)		4 Coring				
Make		Ulterra	HTC	Crossover Sub		1.08		Temp (° C)		5 Circ & Condition	0.50	41.25		
Type		CF713	KM526	21x Heavy Weight DP		197.54		Viscosity (sec)	31	6 Tripping- Bit / Casing		68.50		
IADC Code								PV / YP (cp/lb)	3 / 6	6.1 Tripping- Other		39.25		
Serial Number		39970	5285339					Gels (10s/m)	1 / 2	6.2 Wiper Trip		2.00		
T.F.A. (in)		0.773	0.632					API Filt. (cc)	17.0	6.3 Handle BHA	3.50	61.00		
Depth In (m)		1733	1442					Sand (% Vol)	0.5	7 Service / Slip Line		13.75		
Depth Out (m)		1845	1733					KCl (%)	3.5	8 Repairs		66.75		
Total Meters		112	291					pH (strip)	8.5	10 Survey		2.75		
Hours IADC OnBtm		16.8 11.5	77.5 63.6					LGS %/Vol	5.4	11 Logging		4.00		
ROP IADC OnBtm		6.7 9.7	3.8 4.6					Chlorides (ppm)	18000	12 RU for Csg / Cmt	4.00	30.00		
Condition Out		1 1 CT G X 0 WT TD						Surface Vol. (Bbls)		1021	12.1 Run Casing	11.50	42.50	
FLOW DATA				BHA LENGTH		283.38		Hole Vol.		441	12.2 Cementing		15.25	
RATE (gpm/lps)				BHA WEIGHT		26.76		Mud Mixed		177	13 Wait on Cement		16.00	
AV - DP (fpm/mpm)				STRING WT		19.37		New Hole Drilled			14 BOP's / Wellhead	1.50	19.50	
AV - DC (fpm/mpm)				HOOK LOAD				Downhole Losses			15 Test BOP / Formation	3.00	32.00	
SPP (psi/atm)				WT BELOW JARS				Surface Losses		30	19 Tight hole / Fishing		9.50	
SPP (Calculated)				HOURS: JAR MTR		170.5	77.5	CHEMICAL USAGE						
PUMP DATA				DRAG: UP DOWN							21 Completion			
#1: NAT 8-P-80		STROKE	9.0"	TORQUE: ON OFF							21.1 Well Control			
RATE				SURVEYS: MD INC' AZ'							21.2 Other		40.50	
LINER		6.0"		1827.7 36.2 15.4							TOTALS		24.00	893.25
#2: NAT 8-P-80		STROKE	9.0"								NON PRODUCTIVE		1.25	636.50
RATE											PRODUCTS: USED		REC	ON SITE
LINER		6.0"									DIESEL (L)		3550	47450
#3: IDECO T-800		STROKE	8.5"								BARITE (T)			20.0
RATE											DRILL WATER		130000	70000
LINER		6.0"									DAILY MUD COSTS			
											CUM. MUD COSTS		\$74,812.46	

HOURLY OPERATIONS SUMMARY 0000 to 2400									
From	To	[IADC Code] Description							
0:00	1:30	[12] Continued to rig up for casing, PJSM, picked up landing joint and hanger, landed hanger, noted landing height, laid out hanger.							
1:30	2:00	[12.1] Picked up seal assembly and ran in hole from surface to 44m.							
2:00	3:45	[12.1] PJSM, picked up and ran DDV, installed control lines and pressure tested to 3500 psi for 10 min open and close lines, function tested DDV.							
3:45	10:00	[12.1] Ran in hole with 7" CSG Tie Back assembly and DDV.							
10:00	10:30	[5] Made up wash down joint, broke circulation, tagged TOL, RIH to No Go collar 642.02m, pulled up 3m displaced well to 8.6ppg 2% brine pumped 155bbls at 550GPM.							
10:30	11:30	[12.1] WFT function tested DDV - Ok, Laid out wash down joint, picked up 2 x pup joints for space out, made up 7" CSG hanger.							
11:30	12:30	[12.1] WFT terminated DDV control lines at 7" CSG hanger and press tested to 3000psi - Ok.							
12:30	13:00	[12.1] Ran in hole to TOL at 642.03m, land tie-back 7" CSG with 45,000lbs into well head, top of HGR Lock screws 5.32m from RT.							
13:00	14:30	[14] Cactus well head engineer locked 7" CSG hanger into well head, WFT hooked up DDV control lines and press tested HGR seals 3000psi - Ok.							
14:30	15:00	[12.1] Backed out LH ACME running tool and Laid out, laid out control line sheave.							
15:00	17:30	[15] Pressure tested casing and tie back seal assembly 250 / 2500 psi for 5 / 10 minutes - Ok.							
17:30	18:00	[15] Completed muster drill and function tested ESD's all engines shut down in 6cseconds.							
18:00	20:30	[12] Rigged down casing equipment and rigged up to run in hole with drill out BHA. Flushed poor boy with water through choke manifold.							
20:30	21:45	[6.3] NPT- Inspected BHA on racks, found galled threads and scale blockage in drill collars, changed out drill collars and re-strapped BHA.							
21:45	23:30	[6.3] PJSM, Made up Drill out BHA #17 and ran in hole from surface to 84m.							
23:30	0:00	[6.3] Changed to drill pipe elevators and moved HWDP in derrick.							

HOURLY OPERATIONS SUMMARY 0000 to 0600 on 28/09/18									
0:00	5:30	PJSM, Ran in hole from 84m to 1774, tag cement at 1774m							
5:30	6:00	Drilled out shoe track from 1774m to 1778m, 10k WOB, 400gpm, 1860psi SPP, 60rpm, 300ft.lbs.							
POB-47: Ensign - 18, CTP - 4, OICS - 4, Halliburton - 8, OCS - 2, Site Med - 1, RMN - 1, WFT - 8, Cactus - 1									28,572 Hrs
RECEIVED: Drill water					DISPATCHED:				
MAXIMUM GAS:			BACKGROUND GAS:			CONNECTION GAS:		TRIP GAS:	
SUPERVISOR: Kev Dau / Mark Booth			GEOLOGIST: Phil Allen / Peter Morris			RIG MGR: Gary Hender / Blake Smith			

POB-49: Ensign - 19, CTP - 4, OICS - 4, Halliburton - 10, OCS - 2, Site Med - 1, RMN - 1, WFT - 8, Kinetic - 0				29,736 Hrs
RECEIVED: Stealth equipment			DISPATCHED:	
MAXIMUM GAS:		BACKGROUND GAS:		CONNECTION GAS:
TRIP GAS:				
SUPERVISOR: Kev Dau / Mark Booth		GEOLOGIST: Phil Allen / Peter Morris		RIG MGR: Gary Hender / Adam Delanie

		DAILY DRILLING REPORT				REPORT # 041		30/09/2018				
						SPUD: 21/08/2018		DAY # 087				
WELL	Palm Valley 13	24:00 MD/ TVD	1946 / 1861m BRT	DRILLED	98m	AFE DAYS / COST	46	\$13,467,756				
RIG	ENSIGN 932	FORMATION	Pacoota Sandstone	DAILY COST	\$122,682	CUM.	\$10,979,505					
DAILY OP'S SUMMARY		Unloaded well and drilled 6 1/8" directional production hole from 1845m to 1946m. Down time due to Pason, MWD and mud logging computers.										
FORECAST OP'S		Repair Halliburton computer problems. Drill 6 1/8" directional production hole.										
LAST CASING	178mm	7.0"	SET AT	1842.0m	FIT	12.0 EMW	MAASP	906psi	BOP TEST	26/09	DUE	10/10
LAST L.T.I. 470 Days		SAFETY		1. Discussed operations, JSA's and SOP's, hazard identification, buddy system for new crew. 2. Sunday safety, BBS cards, focus on stop the job, heat stress, operations.				WEATHER: Day		Fine		
								Night		Fine		

BIT INFORMATION				BHA # 18		MUD PROPERTIES		OPERATION		HRS	CUM
WOB(Klb)	2-8	JET V(fps)		TOOL	LENGTH	Mud Type	Stored mud	1 Move/ Rig Up/Down			1122.75
RPM	25	H S I		Bit - PDC	0.21	Depth (m)	1849	2 Drilling	16.00		388.75
BIT NUMBER	16	15		Motor - 4.75	7.80	Density (ppg)	9.10	3 Wash / Ream			24.00
Size (inches)	6.125	6.125		Stabiliser	1.64	ECD (ppg)		4 Coring			
Make	Reed	Smith		MWD	11.08	Temp (° C)		5 Circ & Condition	2.75		50.50
Type	TK63	XRI30D		Float Sub	1.30	Viscosity (sec)	39	6 Tripping- Bit / Casing			84.00
IADC Code		537X		6x Heavy Weight DP	56.34	PV / YP (cp/lb)	10 / 16	6.1 Tripping- Other			44.25
Serial Number	A250214	RJ1906		63x Drill Pipe	581.70	Gels (10s/m)	6 / 7	6.2 Wiper Trip			2.00
T.F.A. (in)	0.902	0.746		Float Sub	0.79	API Filt. (cc)	15.8	6.3 Handle BHA			64.00
Depth In (m)	1848	1845		3x Heavy Weight DP	28.57	Sand (% Vol)	0.1	7 Service / Slip Line	0.50		15.25
Depth Out (m)	IN	1848		5x Drill Collar- 4 3/4"	46.92	KCl (%)	3.5	8 Repairs	4.75		71.50
Total Meters	98	3		Drilling Jars	8.98	pH (strip)	9.5	10 Survey			2.75
Hours IADC OnBtm	16 10.7	1 0.8		17x Heavy Weight DP	161.08	LGS %/Vol	4.3	11 Logging			11.50
ROP IADC OnBtm	6.1 9.2	3 3.7				Chlorides (ppm)	18000	12 RU for Csg / Cmt			30.00
Condition Out	1.1 BT G E 0 No BHA					Surface Vol. (Bbls)	1049	12.1 Run Casing			42.50
FLOW DATA				BHA LENGTH	906.41	Hole Vol.	318	12.2 Cementing			15.25
RATE (gpm/lps)				BHA WEIGHT	102.95	Mud Mixed	352	13 Wait on Cement			16.00
AV - DP (fpm/mpm)				STRING WT	149.38	New Hole Drilled	11.7	14 BOP's / Wellhead			19.50
AV - DC (fpm/mpm)				HOOK LOAD	130.00	Downhole Losses		15 Test BOP / Formation			33.00
SPP (psi/atm)	650	44		WT BELOW JARS	89.17	Surface Losses	406	19 Tight hole / Fishing			9.50
SPP (Calculated)				HOURS: JAR MTR	187.5 77.5	CHEMICAL USAGE		21 Completion			
PUMP DATA				DRAG: UP DOWN	22 5	AMC Defoamer S	11	21.1 Well Control			
#1: NAT 8-P-80	STROKE	9.0"		TORQUE: ON OFF	3000 5000	AMCPacLV	4	21.2 Other			41.00
RATE				SURVEYS: MD INC' AZ'		Caustic Soda	4	TOTALS	24.00	965.25	
LINER	6.0"			1827.7 36.2 15.4		Soda Ash	4	NON PRODUCTIVE	4.75	642.75	
#2: NAT 8-P-80	STROKE	9.0"		1878 35.7 7.5		Xan-Bore	4	PRODUCTS: USED REC ON SITE			
RATE				1906.8 40 2.8		Hydro 327	4	DIESEL (L)	7700		34250
LINER	6.0"			1935.8 40.2 0.5		AMC Superfoam	96	BARITE (T)			20.0
#3: IDECO T-800	STROKE	8.5"						DRILL WATER	10000	60000	120000
RATE								DAILY MUD COSTS			\$7,620.09
LINER	6.0"							CUM. MUD COSTS			\$84,756.38

HOURLY OPERATIONS SUMMARY 0000 to 2400		
From	To	[IADC Code] Description
0:00	1:30	[5] Continued to unload well at 1846mMD, 2000psi to lift fluid, circulated well at 640psi 2700 scfm, mist injection 14 gpm.
1:30	5:15	[2] Directional drill 6 1/8" hole from 1848m to 1871mMD, WOB 2 - 8 Klbs, TQ 5000 ft/lbs @ 15 surface rpm, Air 2700 scfm @ 755 psi, 14 GPM mist rate, ROP 6.13m/hr.
5:15	6:30	[5] Unloaded well circulated to clean well
6:30	8:45	[2] Directional drilled 6 1/8" hole from 1871m to 1894mMD, WOB 2 - 8 Klbs, TQ 5000 ft/lbs @ 15 surface rpm, Air 289 scfm @ 775 psi, 7GPM mist rate, ROP 10.22m/hr.
8:45	10:00	[8] NPT- Worked pipe and circulated while repairing Pason monitoring system connection to Halliburton mud logging unit.
10:00	10:30	[2] Directional drilled 6 1/8" hole from 1894m to 1896mMD, WOB 2 - 8 Klbs, TQ 5000 ft/lbs @ 15 surface rpm, Air 2900 scfm @ 830 psi, 7GPM mist rate, ROP 4m/hr.
10:30	11:30	[8] NPT- Worked pipe and circulated while repairing Pason monitoring system connection to Halliburton mud logging unit.
11:30	13:15	[2] Directional drilled 6 1/8" hole from 1896m to 1903mMD, WOB 2 - 8 Klbs, TQ 2800 ft/lbs @ 35 surface rpm, Air 2400 scfm @ 705 psi, 7GPM mist rate, ROP 4.6m/hr.
13:15	13:45	[8] NPT- Worked pipe and circulated while repairing Pason monitoring system connection to Halliburton mud logging unit. Completed first aid drill.
13:45	15:45	[2] Directional drilled 6 1/8" hole from 1903m to 1920mMD, WOB 2 - 6 Klbs, TQ 2700 ft/lbs @ 35 surface rpm, Air 2500 scfm @ 746 psi, 6GPM mist rate, ROP 8.5m/hr.
15:45	16:15	[7] Serviced rig.
16:15	22:00	[2] Directional drilled 6 1/8" hole from 1920m to 1946mMD, WOB 2 - 6 Klbs, TQ 2700 ft/lbs @ 30 surface rpm, Air 2500 scfm @ 746 psi, 7 GPM mist rate, ROP 4.5 m/hr.
22:00	22:30	[8] NPT- Halliburton MWD lost communication with tool face.
22:30	23:45	[8] NPT- Halliburton mud loggers computer failed / crashed. Flow tested well, 1" orifice plate 15 psi, 0.55 mmcf/d.

HOURLY OPERATIONS SUMMARY 0000 to 0600 on 01/10/18		
From	To	Description
0:00	1:00	Halliburton mud loggers computer failed / crashed.
1:00	2:45	Halliburton mud loggers computer failed / crashed. PJSM and JSA review with WFT and crew, Stepped out JSA and trained crew on RCD change out on live well, Completed AAR.
2:45	3:00	Relogged well section with MWD from 1929 to 1946m, all Halliburton communications / computers repaired.
3:00	6:00	Directional drilled 6 1/8" hole from 1946m to 1953mMD, WOB 2 - 6 Klbs, TQ 2700 ft/lbs @ 30 surface rpm, Air 2500 scfm @ 746 psi, 7 GPM mist rate, ROP 2.3m/hr.

POB-48: Ensign - 19, CTP - 4, OICS - 4, Halliburton - 10, OCS - 1, Site Med - 1, RMN - 1, WFT - 8, Kinetic - 0										30,312 Hrs	
RECEIVED:					DISPATCHED:						
MAXIMUM GAS:			BACKGROUND GAS:			CONNECTION GAS:			TRIP GAS:		
SUPERVISOR: Kev Dau / Mark Booth					GEOLOGIST: Phil Allen / Peter Morris			RIG MGR: Gary Hender / Adam Delanie			

				DAILY DRILLING REPORT				REPORT # 042		1/10/2018			
								SPUD: 21/08/2018		DAY # 088			
WELL	Palm Valley 13	24:00 MD/ TVD	1965 / 1876m BRT	DRILLED	19m	AFE DAYS / COST	46	\$13,467,756					
RIG	ENSIGN 932	FORMATION	Pacoota Sandstone	DAILY COST		\$106,682	CUM.	\$11,086,187					
DAILY OP'S SUMMARY		Repaired Halliburton computer problems. Drilled 6 1/8" directional production hole to 1965mMD. DHM failed, pulled out of hole to above DDV and attempted to close and seal DDV.											
FORECAST OP'S		Close and seal DDV and pull out of hole. Make up BHA and run in hole and open DDV. Run in hole to 1965mMD. Drill 6 1/8" directional production hole.											
LAST CASING		178mm	7.0"	SET AT	1842.0m	FIT	12.0 EMW	MAASP	906psi	BOP TEST	26/09	DUE	10/10
LAST L.T.I. 471 Days		SAFETY		1. Sunday safety, BBS cards for the week, safety alerts, reviewed JSA changing RCD rubber. 2. Discussed tripping, DDV operations, heat stress and housekeeping.						WEATHER: Day Fine Night Fine			

BIT INFORMATION				BHA # 18		MUD PROPERTIES		OPERATION		HRS	CUM
WOB(Klb)	2-12	JET V(fps)		TOOL	LENGTH	Mud Type	Stored mud	1 Move/ Rig Up/Down			1122.75
RPM	25	H S I		Bit - PDC	0.21	Depth (m)	1926	2 Drilling	7.50		396.25
BIT NUMBER	16	15		Motor - 4.75	7.80	Density (ppg)	9.10	3 Wash / Ream			24.00
Size (inches)	6.125	6.125		Stabiliser	1.64	ECD (ppg)		4 Coring			
Make	Reed	Smith		MWD	11.08	Temp (° C)		5 Circ & Condition			50.50
Type	TK63	XRI30D		Float Sub	1.30	Viscosity (sec)	39	6 Tripping- Bit / Casing			84.00
IADC Code		537X		6x Heavy Weight DP	56.34	PV / YP (cp/lb)	10 / 16	6.1 Tripping- Other	13.50		57.75
Serial Number	A250214	RJ1906		63x Drill Pipe	581.70	Gels (10s/m)	6 / 7	6.2 Wiper Trip			2.00
T.F.A. (in)	0.902	0.746		Float Sub	0.79	API Filt. (cc)	15.8	6.3 Handle BHA			64.00
Depth In (m)	1848	1845		3x Heavy Weight DP	28.57	Sand (% Vol)	0.1	7 Service / Slip Line			15.25
Depth Out (m)	1965	1848		5x Drill Collar- 4 3/4"	46.92	KCl (%)	3.5	8 Repairs	3.00		74.50
Total Meters	117	3		Drilling Jars	8.98	pH (strip)	9.5	10 Survey			2.75
Hours IADC OnBtm	23.5 18.2	1 0.8		17x Heavy Weight DP	161.08	LGS %/Vol	4.3	11 Logging			11.50
ROP IADC OnBtm	5 6.4	3 3.7				Chlorides (ppm)	18000	12 RU for Csg / Cmt			30.00
Condition Out						Surface Vol. (Bbls)	1157	12.1 Run Casing			42.50
FLOW DATA				BHA LENGTH	906.41	Hole Vol.	320	12.2 Cementing			15.25
RATE (gpm/lps)				BHA WEIGHT	102.95	Mud Mixed	251	13 Wait on Cement			16.00
AV - DP (fpm/mpm)				STRING WT	150.23	New Hole Drilled	2.3	14 BOP's / Wellhead			19.50
AV - DC (fpm/mpm)				HOOK LOAD	133.00	Downhole Losses		15 Test BOP / Formation			33.00
SPP (psi/atm)	600	41		WT BELOW JARS	89.17	Surface Losses	141	19 Tight hole / Fishing			9.50
SPP (Calculated)				HOURS: JAR MTR	195.0 77.5	CHEMICAL USAGE		21 Completion			
PUMP DATA				DRAG: UP DOWN	22 7	AMC Biocide G	1	21.1 Well Control			
#1: NAT 8-P-80	STROKE	9.0"		TORQUE: ON OFF	3800 2800	AMCPacLV	2	21.2 Other			41.00
RATE				SURVEYS: MD INC AZ			2	TOTALS	24.00	989.25	
LINER	6.0"			1935.8 40.2 0.5		Cautic Soda	2	NON PRODUCTIVE	16.50	659.25	
#2: NAT 8-P-80	STROKE	9.0"				Soda Ash	2	PRODUCTS: USED REC ON SITE			
RATE						Xan-Bore	2	DIESEL (L)	9250	30000	55000
LINER	6.0"					AMC Superfoam	8	BARITE (T)			20.0
#3: IDECO T-800	STROKE	8.5"						DRILL WATER	10000		110000
RATE								DAILY MUD COSTS \$1,005.29			
LINER	6.0"							CUM. MUD COSTS \$85,761.67			

HOURLY OPERATIONS SUMMARY 0000 to 2400		
From	To	[IADC Code] Description
0:00	1:00	[8] NPT- Halliburton mud loggers computer failed / crashed.
1:00	2:45	[8] NPT- Halliburton mud loggers computer failed / crashed. PJSM and JSA review with WFT and crew, Stepped out JSA and trained crew on RCD change out on live well, Completed AAR.
2:45	3:00	[8] NPT- Logged well section with MWD from 1929 to 1946m, all Halliburton communications / computers repaired.
3:00	10:30	[2] Directional drilled 6 1/8" hole from 1946m to 1965mMD, WOB 2 - 6 Klbs, TQ 2700 ft/lbs @ 30 surface rpm, Air 2500 scfm @ 755 psi, 8 GPM mist rate, ROP 2.5m/hr.
10:30	11:00	[6.1] NPT- Had 1000psi press spike on standpipe pressure 755psi to 1780psi, Troubleshoot pressure increase, By-pass air, Held PJSM with crew on tripping pipe from live well.
11:00	11:45	[6.1] NPT- Pulled out of hole to inspect BHA from 1965 to 1802m
11:45	12:15	[6.1] NPT- PJSM, Bled down and removed string float
12:15	13:45	[6.1] NPT- Continued to pull out of hole to inspect BHA from 1802 to 1192m
13:45	13:45	[6.1] NPT- PJSM, Bled down and removed string float
13:45	15:30	[6.1] NPT- Continued to pull out of hole to inspect BHA from 1902 to 659m.
15:30	15:45	[6.1] NPT- PJSM, Bled down and removed string float
15:45	16:30	[6.1] NPT- Continued to pull out of hole to inspect BHA from 659 to 560m (above DDV).
16:30	18:15	[6.1] NPT- PJSM, shut in well and monitored pressure build up, pressure 205 psi, pumped with air to 405 psi.
18:15	18:30	[6.1] NPT- Opened well through choke line and bled down well. Gas bypassing DDV.
18:30	20:00	[6.1] NPT- Shut in well and monitored pressure build up, pressure 240 psi, pumped 3 bbls water to flush DDV. pumped with air to 405 psi.
20:00	20:15	[6.1] NPT- Opened well through choke line and bled down well. Gas bypassing DDV. Suspected DDV stuck in open position.
20:15	21:30	[6.1] NPT- Shut in well and monitored pressure build up, pressure 157 psi, pumped 5 bbls water to flush DDV. pumped with air to 600 psi, cycled DDV open closed.
21:30	22:00	[6.1] NPT- Opened well through choke line and bled down well. Gas bypassing DDV.
22:00	23:00	[6.1] NPT- Shut in well and monitored pressure build up, pressure 115 psi.
23:00	0:00	[6.1] NPT- Opened well through choke line and bled down well. Gas bypassing DDV. Pumped 5 bbls water while venting.

HOURLY OPERATIONS SUMMARY 0000 to 0600 on 02/10/18		
0:00	1:15	Shut in well and monitored pressure build up, pressure 40 psi, pumped 5 bbls water to flush DDV. pumped with air to 400 psi, cycled DDV open closed. Opened well.
1:15	2:00	Shut in well and monitored pressure build up, pressure 0 psi.
2:00	4:15	PJSM, Pulled out of hole with BHA from 560m to 22mMD.
4:15	5:30	PJSM, removed RCD bearing and racked back on stand.
5:30	6:00	Pulled out of hole directional BHA, inspected motor lateral movement and laid out, inspected Bit in gauge no broken cutters can re-run graded 1-1-A-NO-X-in-NO-DTF.

POB-48: Ensign - 19, CTP - 4, OICS - 4, Halliburton - 10, OCS - 1, Site Med - 1, RMN - 1, WFT - 8, Kinetic - 0									30,888 Hrs		
RECEIVED: XT39 DP and 4 3/4" DHM						DISPATCHED: Subs and 6" DHM					
MAXIMUM GAS:			11.1 U @ 1953m			BACKGROUND GAS:			10 U		
CONNECTION GAS:						TRIP GAS:			33.87 U		
SUPERVISOR: Kev Dau / Mark Booth						GEOLOGIST: Phil Allen / Peter Morris			RIG MGR: Gary Hender / Adam Delanie		

			DAILY DRILLING REPORT			REPORT # 043		2/10/2018	
						SPUD: 21/08/2018		DAY # 089	
WELL	Palm Valley 13	24:00 MD/ TVD	1965 / 1876m BRT	DRILLED		AFE DAYS / COST	46	\$13,467,756	
RIG	ENSIGN 932	FORMATION	Pacoota Sandstone		DAILY COST	\$106,682	CUM.	\$11,192,869	
DAILY OP'S SUMMARY		Closed and sealed DDV and pulled out of hole. Pressure tested annular. Made up BHA #19 and ran in hole and opened DDV. Ran in hole to 1965mMD. Repaired TDS hydraulics. Changed out RCD rubber. Circulated well.							
FORECAST OP'S		Drill 6 1/8" directional production hole.							
LAST CASING		178mm	7.0"	SET AT	1842.0m	FIT	12.0 EMW	MAASP	906psi
LAST L.T.I. 472 Days		SAFETY		1. Discussed BBS cards, bullying policy, operations and snakes about. 2. Discussed operations, handling BHA, hazards on working over live well and PPE.				WEATHER: Day Fine Night Fine	

BIT INFORMATION			BHA # 19		MUD PROPERTIES		OPERATION	HRS	CUM	
WOB(Klb)		JET V(fps)		TOOL	LENGTH	Mud Type	1 Move/ Rig Up/Down		1122.75	
RPM		H S I		Bit - PDC	0.21	Depth (m)	2 Drilling		396.25	
BIT NUMBER	16	15		Motor - 4.75	7.49	Density (ppg)	3 Wash / Ream		24.00	
Size (inches)	6.125	6.125		Stabiliser	1.64	ECD (ppg)	4 Coring			
Make	Reed	Smith		MWD	11.08	Temp (° C)	5 Circ & Condition	3.00	53.50	
Type	TK63	XRI30D		Float Sub	0.84	Viscosity (sec)	6 Tripping- Bit / Casing		84.00	
IADC Code		537X		6x Heavy Weight DP	58.55	PV / YP (cp/lb)	6.1 Tripping- Other	13.00	70.75	
Serial Number	A250214	RJ1906		62x Drill Pipe	580.11	Gels (10s/m)	6.2 Wiper Trip		2.00	
T.F.A. (in)	0.902	0.746		Float Sub	0.79	API Filt. (cc)	6.3 Handle BHA	6.25	70.25	
Depth In (m)	1848	1845		3x Heavy Weight DP	28.56	Sand (% Vol)	7 Service / Slip Line		15.25	
Depth Out (m)	1965	1848		5x Drill Collar- 4 3/4"	46.92	KCl (%)	8 Repairs	0.50	75.00	
Total Meters	117	3		Drilling Jars	8.98	pH (strip)	10 Survey		2.75	
Hours IADC OnBtm	23.5 16.1	1 0.8		17x Heavy Weight DP	160.93	LGS %/Vol	11 Logging		11.50	
ROP IADC OnBtm	5 7.3	3 3.7				Chlorides (ppm)	12 RU for Csg / Cmt		30.00	
Condition Out	1 1 No A X 0 No DTF					Surface Vol. (Bbls)	12.1 Run Casing		42.50	
FLOW DATA			BHA LENGTH	906.10		Hole Vol.	12.2 Cementing		15.25	
RATE (gpm/lps)			BHA WEIGHT	102.81		Mud Mixed	13 Wait on Cement		16.00	
AV - DP (fpm/mpm)			STRING WT	150.10		New Hole Drilled	14 BOP's / Wellhead		19.50	
AV - DC (fpm/mpm)			HOOK LOAD			Downhole Losses	15 Test BOP / Formation	1.25	34.25	
SPP (psi/atm)			WT BELOW JARS	89.04		Surface Losses	19 Tight hole / Fishing		9.50	
SPP (Calculated)			HOURS: JAR MTR	195.0	77.5	CHEMICAL USAGE	21 Completion			
PUMP DATA			DRAG: UP DOWN			AMC Superfoam	1			
#1: NAT 8-P-80	STROKE	9.0"	TORQUE: ON OFF							
RATE			SURVEYS: MD INC AZ							
LINER	6.0"		1935.8 40.2 0.5							
#2: NAT 8-P-80	STROKE	9.0"								
RATE										
LINER	6.0"									
#3: IDECO T-800	STROKE	8.5"								
RATE										
LINER	6.0"									
HOURLY OPERATIONS SUMMARY 0000 to 2400										
From	To	[IADC Code] Description								
0:00	1:15	[6.1] NPT- Shut in well and monitored pressure build up, pressure 40 psi, pumped 5 bbls water to flush DDV. pumped with air to 400 psi, cycled DDV open closed. Opened well.								
1:15	2:00	[6.1] NPT- Shut in well and monitored pressure build up, pressure 0 psi.								
2:00	4:15	[6.1] NPT- PJSM, Pulled out of hole with BHA from 560m to 22mMD.								
4:15	5:30	[6.1] NPT- PJSM, removed RCD bearing and racked back on stand.								
5:30	7:30	[6.1] NPT- Pulled out of hole directional BHA, inspected motor lateral movement and laid out, inspected Bit in gauge no broken cutters can re-run graded 1-1-A-NO-X-In-NO-DTF.								
7:30	8:45	[15] NPT- Made up cup tester and ran in hole, pressure test annular 250/2500 5/10 min. - OK, laid out cup tester.								
8:45	10:15	[6.3] NPT- Picked up and tested DHM from BHA #18 - not suitable for rerun, Laid out DHM.								
10:15	15:00	[6.3] NPT- Picked up and made up BHA #19 and ran in hole from surface to 516m. Shallow hole test DHM / MWD - ok.								
15:00	16:00	[6.1] NPT- Shut in well and pressured up to 570 psi, opened DDV and flowed well through choke line.								
16:00	18:30	[6.1] NPT- Ran in hole with directional BHA from 516m to 1225mMD.								
18:30	19:00	[6.1] NPT- Serviced rig and troubleshoot TDS torque problem.								
19:00	19:30	[8] NPT- Continued to trouble shoot TDS torque problem.								
19:30	20:00	[5] NPT- Broke circulation and circulated well. RCD rubber leaking.								
20:00	22:00	[5] NPT- PJSM, Changed out RCD rubber.								
22:00	23:30	[6.1] NPT- Ran in hole with directional BHA from 1225m to 1948mMD.								
23:30	0:00	[5] Circulated well and washed to 1965mMD.								
HOURLY OPERATIONS SUMMARY 0000 to 0600 on 03/10/18										
0:00	2:45	Directional drilled 6 1/8" hole from 1965m to 1979mMD, WOB 2 - 6 Klbs, TQ 6000 ft/lbs @ 25 surface rpm, Air 2500 scfm @ 650 psi, 7 GPM mist rate, ROP 5 m/hr.								
2:45	3:30	Pressure spike to 1400 psi, suspected stalled DHM, unable to bleed down string with string floats in drill string.								
3:30	4:30	Pull out of hole from 1879m to 1802mMD, bleed down string float and laid out.								
4:30	5:00	Circulated string at 2100 cfm 510 psi.								
5:00	5:30	Ran in hole from 1802m to 1979mMD.								
5:30	6:00	Break circulation and circulate well.								
POB-48: Ensign - 19, CTP - 4, OICS - 4, Halliburton - 10, OCS - 1, Site Med - 1, RMN - 1, WFT - 8, Kinetic - 0								31,464 Hrs		
RECEIVED:					DISPATCHED:					
MAXIMUM GAS:			BACKGROUND GAS:			CONNECTION GAS:			TRIP GAS:	
SUPERVISOR: Kev Dau / Mark Booth			GEOLOGIST: Phil Allen / Peter Morris			RIG MGR: Gary Hender / Adam Delanie				

			DAILY DRILLING REPORT				REPORT # 044		3/10/2018			
							SPUD: 21/08/2018		DAY # 090			
WELL	Palm Valley 13	24:00 MD/ TVD	1984 / 1889m BRT	DRILLED	19m	AFE DAYS / COST	46	\$13,467,756				
RIG	ENSIGN 932	FORMATION	Pacoota Sandstone	DAILY COST	\$122,682	CUM.	\$11,315,551					
DAILY OP'S SUMMARY		Drilled 6 1/8" directional production hole. Pressure spike trapped pressure in drill string, pulled back to string float and bled of pressure. Circulated, ran in hole. Drilled 6 1/8" directional production hole from 1879m to 1984mMD. Pressure spike trapped pressure in drill string, unable to release pressure. Pulled out of hole to surface.										
FORECAST OP'S		Make up directional BHA and run in hole to 550m, open DDV and continue to run in hole. Drill 6 1/8" directional hole.										
LAST CASING	178mm	7.0"	SET AT	1842.0m	FIT	12.0 EMW	MAASP	906psi	BOP TEST	26/09	DUE	10/10
LAST L.T.I. 473 Days		SAFETY		1. Discussed, BBS cards, SOP's & JSA's, hydration and flash safety alert. 2. Discussed anyone can stop the job, JSA reviews, well control equipment and operations.				WEATHER: Day		Fine		
								Night		Fine		

BIT INFORMATION			BHA # 19		MUD PROPERTIES		OPERATION	HRS	CUM
WOB(Klb)	2-8	JET V(fps)	TOOL	LENGTH	Mud Type	Stored mud	1 Move/ Rig Up/Down		1122.75
RPM	30	H S I	Bit - PDC	0.21	Depth (m)	1979	2 Drilling	5.00	401.25
BIT NUMBER	17 RR16	16	Motor - 4.75	7.49	Density (ppg)	9.10	3 Wash / Ream		24.00
Size (inches)	6.125	6.125	Stabiliser	1.64	ECD (ppg)		4 Coring		
Make	Reed	Reed	MWD	11.08	Temp (° C)		5 Circ & Condition	1.25	54.75
Type	TK63	TK63	Float Sub	0.84	Viscosity (sec)	39	6 Tripping- Bit / Casing		84.00
IADC Code			6x Heavy Weight DP	58.55	PV / YP (cp/lb)	10 / 16	6.1 Tripping- Other	12.25	83.00
Serial Number	A250214	A250214	62x Drill Pipe	580.11	Gels (10s/m)	6 / 7	6.2 Wiper Trip		2.00
T.F.A. (in)	0.902	0.902	Float Sub	0.79	API Filt. (cc)	15.8	6.3 Handle BHA	5.50	75.75
Depth In (m)	1965	1848	3x Heavy Weight DP	28.56	Sand (% Vol)	0.1	7 Service / Slip Line		15.25
Depth Out (m)	1984	1965	5x Drill Collar- 4 3/4"	46.92	KCl (%)	3.5	8 Repairs		75.00
Total Meters	19	117	Drilling Jars	8.98	pH (strip)	9.5	10 Survey		2.75
Hours IADC OnBtm	5 3.1	23.5 16.1	17x Heavy Weight DP	160.93	LGS %/Vol	4.3	11 Logging		11.50
ROP IADC OnBtm	3.8 6.1	5 7.3			Chlorides (ppm)	18000	12 RU for Csg / Cmt		30.00
Condition Out	1 1 No A X 0 No DTF				Surface Vol. (Bbls)	1126	12.1 Run Casing		42.50
FLOW DATA			BHA LENGTH	906.10	Hole Vol.	322	12.2 Cementing		15.25
RATE (gpm/lps)			BHA WEIGHT	102.81	Mud Mixed	118	13 Wait on Cement		16.00
AV - DP (fpm/mpm)			STRING WT	150.95	New Hole Drilled	2.3	14 BOP's / Wellhead		19.50
AV - DC (fpm/mpm)			HOOK LOAD	135.00	Downhole Losses		15 Test BOP / Formation		34.25
SPP (psi/atm)	650	44	WT BELOW JARS	89.04	Surface Losses	147	19 Tight hole / Fishing		9.50
SPP (Calculated)			HOURS: JAR MTR	200.0 77.5	CHEMICAL USAGE		21 Completion		
PUMP DATA			DRAG: UP DOWN	20 10	AMC Biocide G	1	21.1 Well Control		
#1: NAT 8-P-80	STROKE	9.0"	TORQUE: ON OFF	3000 6000	AMCPacLV	1	21.2 Other		41.00
RATE			SURVEYS: MD INC' AZ'		Soda Ash	1	TOTALS	24.00	1037.25
LINER	6.0"		1935.8 40.2 0.5		Xan-Bore	1	NON PRODUCTIVE	19.00	701.75
#2: NAT 8-P-80	STROKE	9.0"	1966.5 44.1 0.1		Hydro 327	1	PRODUCTS: USED REC ON SITE		
RATE							DIESEL (L)	4000	51000
LINER	6.0"						BARITE (T)		20.0
#3: IDECO T-800	STROKE	8.5"					DRILL WATER	60000	190000
RATE							DAILY MUD COSTS \$311.62		
LINER	6.0"						CUM. MUD COSTS \$82,349.77		

HOURLY OPERATIONS SUMMARY 0000 to 2400		
From	To	[IADC Code] Description
0:00	2:45	[2] Directional drilled 6 1/8" hole from 1965m to 1979mMD, WOB 2 - 6 Klbs, TQ 6000 ft/lbs @ 25 surface rpm, Air 2500 scfm @ 650 psi, 7 GPM mist rate, ROP 5 m/hr.
2:45	3:30	[5] NPT- Pressure spike to 1400 psi, suspected stalled DHM, unable to bleed down string with string floats in drill string.
3:30	4:30	[6.1] NPT- Pull out of hole from 1879m to 1802mMD, bled down string float and laid out.
4:30	5:00	[5] NPT- Circulated string at 2100 cfm 510 psi.
5:00	5:30	[6.1] NPT- Ran in hole from 1802m to 1979mMD.
5:30	7:45	[2] Directional drilled 6 1/8" hole from 1979m to 1984mMD, WOB 2 - 6 Klbs, TQ 5000 ft/lbs @ 30 surface rpm, Air 2500 scfm @ 650 psi, 10 GPM mist rate, ROP m/hr.
7:45	8:45	[6.1] NPT- Pressure spike to 850 psi, suspected stalled DHM, unable to bleed down string with string floats in drill string.
8:45	10:45	[6.1] NPT- Pulled back to 1976mMD, bled down drill string and removed float, attempted to circulate string - no circulation.
10:45	12:00	[6.1] NPT- Pulled out of hole from 1976m to 1424mMD to inspect string floats.
12:00	15:30	[6.1] NPT- Pulled out of hole from 1424m to 560mMD, inspected and bled down drill string floats and laid out.
15:30	16:30	[6.1] NPT- PJSM, shut in well and pressured up to 670 psi and closed DDV.
16:30	17:30	[6.1] NPT- Bled down well through choke line, shut in and monitored well.
17:30	18:30	[6.1] NPT- Pulled out of hole from 560m to 22mMD.
18:30	19:15	[6.3] NPT- PJSM, removed RCD and racked back on stand.
19:15	20:45	[6.3] NPT- Broke out and inspected directional BHA.
20:45	23:30	[6.3] NPT- Made up BHA #20 with TCI bit and changed out dart type for flapper type float valves. Shallow hole tested DHM.
23:30	0:00	[6.3] NPT- PJSM, installed RCD.

HOURLY OPERATIONS SUMMARY 0000 to 0600 on 04/10/18		
From	To	Description
0:00	2:45	Tripped in hole from 21m to 563m., drifting tubulars.
2:45	3:45	Made up Top Drive. Shallow tested motor. Closed annular BOP and pressured up against DDV to 686psi. Opened DDV. Bled off well through choke to flare pit.
3:45	6:00	Tripped in hole from 563m to 750m, drifting tubulars.

POB-49: Ensign - 19, CTP - 5, OICS - 4, Halliburton - 10, OCS - 1, Site Med - 1, RMN - 1, WFT - 8, Kinetic - 0								32,052 Hrs		
RECEIVED:					DISPATCHED:					
MAXIMUM GAS:			BACKGROUND GAS:			CONNECTION GAS:			TRIP GAS:	
SUPERVISOR: Kev Dau / Mark Booth/Dave York			GEOLOGIST: Phil Allen / Peter Morris			RIG MGR: Gary Hender / Adam Delanie				

			DAILY DRILLING REPORT				REPORT # 045		4/10/2018			
							SPUD: 21/08/2018		DAY # 091			
WELL	Palm Valley 13	24:00 MD/ TVD	2012 / 1909m BRT	DRILLED	28m	AFE DAYS / COST	46	\$13,467,756				
RIG	ENSIGN 932	FORMATION	Pacoota Sandstone		DAILY COST	\$106,682	CUM.	\$11,422,233				
DAILY OP'S SUMMARY		Tripped in the hole from 21m to 1984.7m, with rig service and rig repair. Directionally drilled 6 1/8" hole from 1984.7m to 2012m. Flow tested well. Worked string, establishing drilling parameters.										
FORECAST OP'S		Directionally drill 6 1/8" hole.										
LAST CASING	178mm	7.0"	SET AT	1842.0m	FIT	12.0 EMW	MAASP	906psi	BOP TEST	26/09	DUE	10/10
LAST L.T.I. 474 Days		SAFETY		1. Drifting pipe in mast. Trip in UBD. Crew training. 2. Trip in drifting pipe. Familiarise incoming crew.				WEATHER: Day		Fine		
								Night		Fine		

BIT INFORMATION			BHA # 19		MUD PROPERTIES		OPERATION	HRS	CUM	
WOB(Klb)	15	JET V(fps)		TOOL	LENGTH	Mud Type	Stored mud	1 Move/ Rig Up/Down		1122.75
RPM	40	H S I		Bit - PDC	0.21	Depth (m)	1979	2 Drilling	4.50	405.75
BIT NUMBER	18	17 RR16		Motor - 4.75	7.49	Density (ppg)	9.10	3 Wash / Ream	0.25	24.25
Size (inches)	6.125	6.125		Stabiliser	1.64	ECD (ppg)		4 Coring		
Make	Smith	Reed		MWD	11.08	Temp (° C)		5 Circ & Condition	2.50	57.25
Type	XR400D	TK63		Float Sub	0.84	Viscosity (sec)	39	6 Tripping- Bit / Casing	1.50	85.50
IADC Code	617			6x Heavy Weight DP	58.55	PV / YP (cp/lb)	10 / 16	6.1 Tripping- Other	10.50	93.50
Serial Number	RJ3288	A250214		62x Drill Pipe	580.11	Gels (10s/m)	6 / 7	6.2 Wiper Trip		2.00
T.F.A. (in)	0.746	0.902		Float Sub	0.79	API Filt. (cc)	15.8	6.3 Handle BHA		75.75
Depth In (m)	1984	1965		3x Heavy Weight DP	28.56	Sand (% Vol)	tr	7 Service / Slip Line	0.50	15.75
Depth Out (m)	IN	1984		5x Drill Collar- 4 3/4"	46.92	KCl (%)	3.5	8 Repairs	2.00	77.00
Total Meters	28	19		Drilling Jars	8.98	pH (strip)	9.5	10 Survey		2.75
Hours IADC OnBtm	4.5 3.6	5 3.1		17x Heavy Weight DP	160.93	LGS %/Vol	4.3	11 Logging		11.50
ROP IADC OnBtm	6.2 7.8	3.8 6.1				Chlorides (ppm)	18000	12 RU for Csg / Cmt		30.00
Condition Out	1.1 No A X 0 No DTF					Surface Vol. (Bbls)		12.1 Run Casing		42.50
FLOW DATA				BHA LENGTH	906.10	Hole Vol.	325	12.2 Cementing		15.25
RATE (gpm/lps)				BHA WEIGHT	102.81	Mud Mixed		13 Wait on Cement		16.00
AV - DP (fpm/mpm)				STRING WT	152.20	New Hole Drilled	3.3	14 BOP's / Wellhead		19.50
AV - DC (fpm/mpm)				HOOK LOAD	140.00	Downhole Losses		15 Test BOP / Formation		34.25
SPP (psi/atm)	600	41		WT BELOW JARS	89.04	Surface Losses	1123	19 Tight hole / Fishing		9.50
SPP (Calculated)				HOURS: JAR MTR	204.5 77.5	CHEMICAL USAGE		21 Completion		
PUMP DATA				DRAG: UP DOWN	15 12	AMC Biocide G	1	21.1 Well Control		
#1: NAT 8-P-80	STROKE	9.0"		TORQUE: ON OFF	7000 5500	AMCPacLV	1	21.2 Other	2.25	43.25
RATE				SURVEYS: MD INC' AZ'		Caustic Soda	1	TOTALS	24.00	1061.25
LINER	6.0"			1966.5 44.1 0.1		Soda Ash	1	NON PRODUCTIVE	14.00	715.75
#2: NAT 8-P-80	STROKE	9.0"				Xan-Bore	1	PRODUCTS: USED REC ON SITE		
RATE						Hydro 327	1	DIESEL (L)		
LINER	6.0"							BARITE (T)		20.0
#3: IDECO T-800	STROKE	8.5"						DRILL WATER		
RATE								DAILY MUD COSTS \$365.30		
LINER	6.0"							CUM. MUD COSTS \$82,715.07		

HOURLY OPERATIONS SUMMARY 0000 to 2400		
From	To	[IADC Code] Description
0:00	2:45	[6.1] NPT- Tripped in hole from 21m to 563m., drifting tubulars.
2:45	3:45	[21.2] Made up Top Drive. Shallow tested motor. Closed annular BOP and pressured up against DDV to 686psi. Opened DDV. Bled off well through choke to flare pit.
3:45	10:00	[6.1] NPT- Tripped in hole from 563m to 1817m, drifting tubulars.
10:00	10:30	[7] Serviced rig.
10:30	12:00	[6.1] NPT- Changed out RCD bearing assembly.
12:00	14:00	[8] NPT- Rig repair. Pason comms failure.
14:00	15:00	[6] NPT- Tripped in hole from 1817m to 1950m, drifting tubulars.
15:00	15:30	[6] NPT- Made up string float and established parameters.
15:30	15:45	[3] Washed down from 1950m to tag bottom at 1984.7m.
15:45	20:15	[2] Directionally drilled 1984.7m to 2012m. 15klbs WOB. Air: 2300SCFM at 600psi. 8GPM mist injection. 40surface RPM. Average ROP: 6m/hr.
20:15	20:45	[5] Standpipe pressure increase. Increased flare. Observe pressures.
20:45	22:00	[21.2] Shut in well and conduct flow test. 44psi through 2" orifice = 4.6MMcfd according to Weatherford information.
22:00	0:00	[5] Worked string attempting to establish drilling parameters due to high circulating pressure 1700psi+

HOURLY OPERATIONS SUMMARY 0000 to 0600 on 05/10/18		
0:00	1:15	Directionally drilled 6 1/8" hole from 2012 to 2012.7m, sliding.
1:15	2:45	Unloaded well, loaded up with foam / water.
2:45	4:15	Established drilling rates for air and mist.
4:15	6:00	Directionally drilled 6 1/8" hole from 2012.7 to 2015.5m. Air: 2350SCFM at 1650psi. 17GPM mist injection rate. Av ROP: 1.6m/hr.

POB-48: Ensign - 19, CTP - 4, OICS - 4, Halliburton - 10, OCS - 1, Site Med - 1, RMN - 1, WFT - 8, Kinetic - 0								32,628 Hrs		
RECEIVED:						DISPATCHED:				
MAXIMUM GAS:			BACKGROUND GAS:			CONNECTION GAS:			TRIP GAS:	
SUPERVISOR: Kev Dau / Dave York			GEOLOGIST: Phil Allen / Peter Morris			RIG MGR: Gary Hender / Adam Delanie				

POB-49: Ensign - 20, CTP - 4, OICS - 4, Halliburton - 10, OCS - 1, Site Med - 1, RMN - 1, WFT - 8				33,228 Hrs
RECEIVED:		DISPATCHED:		
MAXIMUM GAS:		BACKGROUND GAS:		CONNECTION GAS:
TRIP GAS:				
SUPERVISOR: Kev Dau / Dave York		GEOLOGIST: Phil Allen / Peter Morris		RIG MGR: Gary Hender / Adam Delanie

			DAILY DRILLING REPORT			REPORT # 047		6/10/2018	
						SPUD: 21/08/2018		DAY # 093	
WELL	Palm Valley 13	24:00 MD/ TVD	2022 / 1914m BRT	DRILLED		AFE DAYS / COST	46	\$13,467,756	
RIG	ENSIGN 932	FORMATION	Pacoota Sandstone		DAILY COST	\$142,682	CUM.	\$11,671,597	
DAILY OP'S SUMMARY		Bled down well and opened up well to blooie line. Tripped out from 1830m to 570m. Closed DDV. Tripped out to surface. Changed BHA and Bit. Tripped in to 570m. Opened DDV. Bled down well and tripped in to 1847m, having changed out the RCD bearing at 1130m.							
FORECAST OP'S		Directionally drill 6 1/8" hole.							
LAST CASING		178mm	7.0"	SET AT	1842.0m	FIT	12.0 EMW	MAASP	906psi
						BOP TEST	26/09	DUE 10/10	
LAST L.T.I. 476 Days		SAFETY				1. Bleeding down well. Tripping. Tripping through RCD. 2. Tripping. Kick drill performed. Close DDV.		WEATHER: Day Fine Night Fine	

BIT INFORMATION			BHA # 20		MUD PROPERTIES		OPERATION	HRS	CUM
WOB(Klb)		JET V(fps)		TOOL		Stored mud	1 Move/ Rig Up/Down		1122.75
RPM		H S I		Bit - PDC	0.23	2022	2 Drilling		417.00
BIT NUMBER	18	17 RR16		Motor - 4.75	7.67	9.10	3 Wash / Ream		24.25
Size (inches)	6.125	6.125		Float Sub	0.70		4 Coring		
Make	Smith	Reed		Stabiliser	1.64		5 Circ & Condition		60.75
Type	XR400D	TK63		MWD	11.08	39	6 Tripping- Bit / Casing	16.50	104.75
IADC Code	617			Float Sub	1.88	10 / 16	6.1 Tripping- Other		93.75
Serial Number	RJ3288	A250214		Crossover Sub	1.64	6 / 7	6.2 Wiper Trip		2.00
T.F.A. (in)	0.746	0.902		42x Drill Pipe	404.52	15.8	6.3 Handle BHA	2.50	78.25
Depth In (m)	1984	1965		6x Heavy Weight DP	56.86	tr	7 Service / Slip Line	0.50	16.25
Depth Out (m)	2022	1984		4x Drill Collar- 4 3/4"	37.51	3.5	8 Repairs		77.00
Total Meters	38	19		Drilling Jars	8.98	9.5	10 Survey		2.75
Hours IADC OnBtm	15.8 12.5	5 3.1		Drill Collar- 4 3/4"	9.41	4.3	11 Logging		11.50
ROP IADC OnBtm	2.4 3.1	3.8 6.1		Crossover Sub	1.08	18000	12 RU for Csg / Cmt		30.00
Condition Out	2 2 WT A F 1 SD PR			20x Heavy Weight DP	188.05		12.1 Run Casing		42.50
FLOW DATA			BHA LENGTH	731.25		Surface Vol. (Bbls)	1134		
RATE (gpm/lps)			BHA WEIGHT	103.05		Hole Vol.	322		
AV - DP (fpm/mpm)			STRING WT	160.71		Mud Mixed			
AV - DC (fpm/mpm)			HOOK LOAD	130.00		New Hole Drilled			
SPP (psi/atm)			WT BELOW JARS	69.32		Downhole Losses			
SPP (Calculated)			HOURS: JAR MTR	215.5	77.5	Surface Losses	62		
PUMP DATA			DRAG: UP DOWN			CHEMICAL USAGE			
#1: NAT 8-P-80	STROKE	9.0"	TORQUE: ON OFF						
RATE			SURVEYS: MD INC" AZ"						
LINER	6.0"		1997 48 1.9						
#2: NAT 8-P-80	STROKE	9.0"							
RATE									
LINER	6.0"								
#3: IDECO T-800	STROKE	8.5"							
RATE									
LINER	6.0"								
							PRODUCTS: USED REC ON SITE		
							DIESEL (L)	6950	30000
							BARITE (T)		20.0
							DRILL WATER	30000	70000
							DAILY MUD COSTS		
							CUM. MUD COSTS \$87,543.72		

HOURLY OPERATIONS SUMMARY 0000 to 2400		
From	To	[IADC Code] Description
0:00	2:30	[21.2] Bled down well through choke manifold to flare pit with pressure stabilising at 130psi through centre line of manifold. Equalised pressure above pipe rams to 130psi. Opened rams. Closed manual 7 1/16 blooie line valve. Opened hydraulic 7 1/16 blooie line valve. Opened manual valve.
2:30	7:00	[6] Tripped out of hole to 530m, 70m above DDV.
7:00	8:30	[6] Closed DDV as per WFT SOP's.
8:30	10:30	[6] Pulled out of hole from 530m to 60m with primary jet on 600scfm.
10:30	11:00	[6] PJSM, PTW, Removed RCD as per WFT/Ensign SOP's
11:00	12:00	[6] Pulled out of hole from 60m to surface with primary jet on 600scfm, Laid out bent and damaged mud motor, break out bit.
12:00	14:30	[6.3] Picked up replacement motor, adjusted bend to 1.5 degrees. Made up new bit, changed out MWD probe and scribed tool face.
14:30	15:00	[7] Serviced rig.
15:00	18:00	[6] Tripped in hole to 570m, reconfiguring BHA, as required.
18:00	18:30	[6] Installed RCD bearing assembly.
18:30	19:30	[21.2] Pressured up well and opened DDV at 700psi. Bled down well through choke manifold to 150psi, opened up well to blooie line. Equalised pressure and opened annular preventer.
19:30	21:15	[6] Tripped in hole to 1130m. RCD packer leaking.
21:15	22:15	[21.2] Changed out RCD bearing assembly.
22:15	0:00	[6] Tripped in hole to tag up at 1847m. Installed float at 1843m.

HOURLY OPERATIONS SUMMARY 0000 to 0600 on 07/10/18		
From	To	Description
0:00	0:45	Washed through 1843m to 1862m.
0:45	1:15	Tripped in 1862m to 1949m.
1:15	2:30	Washed and reamed from 1949m to 2022m. Light reaming at 1986m to 1987m. Less than 1m fill.
2:30	4:15	Directionally drilled 6 1/8" hole, sliding, from 2022m to 2026m. Air: 2000scfm at 1040psi, Mist: 20gpm, WOB: 5 - 7klbs. Av ROP: 2.3m/hr.
4:15	6:00	Directionally drilled 6 1/8" hole, rotating, from 2026m to 2036m. Air: 2000scfm at 1040psi, Mist: 20gpm, WOB: 5 klbs. Surface RPM: 30. Torque: 7000ft lbs. Av ROP: 5.7m/hr.

POB-50: Ensign - 20, CTP - 4, OICS - 4, Halliburton - 10, OCS - 1, Site Med - 1, RMN - 1, WFT - 8, Pason - 1								33,828 Hrs		
RECEIVED:					DISPATCHED:					
MAXIMUM GAS:			BACKGROUND GAS:			CONNECTION GAS:			TRIP GAS:	
SUPERVISOR: Kev Dau / Dave York			GEOLOGIST: Phil Allen / Peter Morris			RIG MGR: Gary Hender / Adam Delanie				

POB-51: Ensign - 20, CTP - 4, OICS - 4, Halliburton - 10, OCS - 1, Site Med - 1, RMN - 1, WFT - 9, Pason - 1				34,440 Hrs
RECEIVED:			DISPATCHED:	
MAXIMUM GAS:		BACKGROUND GAS:		CONNECTION GAS:
SUPERVISOR: Kev Dau / Dave York		GEOLOGIST: Phil Allen / Peter Morris		RIG MGR: Gary Hender / Adam Delanie

			DAILY DRILLING REPORT			REPORT # 051		10/10/2018				
						SPUD: 21/08/2018		DAY # 097				
WELL	Palm Valley 13	24:00 MD/ TVD	2133 / 1967m BRT	DRILLED		AFE DAYS / COST	46	\$13,467,756				
RIG	ENSIGN 932	FORMATION	Pacoota Sandstone		DAILY COST	\$106,682	CUM.	\$12,098,325				
DAILY OP'S SUMMARY		Pressure tested BOPE. Serviced rig and slipped and cut drilling line. Changed out spare RCD element. Pressure tested kelly cocks. Made up directional BHA.										
FORECAST OP'S		Trip in the hole to casing shoe. Wash to bottom. Directionally drill 6 1/8" hole.										
LAST CASING	178mm	7.0"	SET AT	1842.0m	FIT	12.0 EMW	MAASP	906psi	BOP TEST	10/10	DUE	24/10
LAST L.T.I. 480 Days		SAFETY					1. PTSM: Pressure testing. Reviewed BBS cards. Housekeeping. 2. PTSM: Pressure testing. Cleaning mud tanks. Water intake.		WEATHER: Day Overcast Night Overcast			

BIT INFORMATION			BHA # 21		MUD PROPERTIES		OPERATION	HRS	CUM	
WOB(Klb)		JET V(fps)		TOOL	LENGTH	Mud Type	Stored mud	1 Move/ Rig Up/Down		1122.75
RPM		H S I		Bit - PDC	0.23	Depth (m)	2120	2 Drilling		469.75
BIT NUMBER	19		18	Motor - 4.75	7.67	Density (ppg)	9.10	3 Wash / Ream		26.25
Size (inches)	6.125	6.125		Float Sub	0.70	ECD (ppg)		4 Coring		
Make	Ulterra	Smith		Stabiliser	1.64	Temp (° C)		5 Circ & Condition		62.00
Type	U71M-DEGU	XR400D		MWD	11.08	Viscosity (sec)	39	6 Tripping- Bit / Casing		105.25
IADC Code		617		Float Sub	1.88	PV / YP (cp/lb)	10 / 16	6.1 Tripping- Other		104.25
Serial Number	41000	RJ3288		Crossover Sub	0.82	Gels (10s/m)	6 / 7	6.2 Wiper Trip		2.00
T.F.A. (in)	0.994	0.746		43x Drill Pipe	404.52	API Filt. (cc)	15.8	6.3 Handle BHA	2.00	83.50
Depth In (m)	2022	1984		Crossover Sub	0.82	Sand (% Vol)	tr	7 Service / Slip Line	2.00	18.75
Depth Out (m)	2133	2022		14x Heavy Weight DP	132.64	KCl (%)	3.5	8 Repairs		77.00
Total Meters	111	38		Crossover Sub	0.36	pH (strip)	9.5	10 Survey		2.75
Hours IADC OnBtm	52.8 44.3	15.8 12.5		Drilling Jars	8.98	LGS %/Vol	4.3	11 Logging		11.50
ROP IADC OnBtm	2.1 2.5	2.4 3.1		Crossover Sub	1.08	Chlorides (ppm)	18000	12 RU for Csg / Cmt		30.00
Condition Out	2 1 WT S X 0 No BHA			11x Heavy Weight DP	103.50	Surface Vol. (Bbls)	1167	12.1 Run Casing		42.50
FLOW DATA				BHA LENGTH	675.92	Hole Vol.	#REF!	12.2 Cementing		15.25
RATE (gpm/lps)				BHA WEIGHT	#REF!	Mud Mixed	103	13 Wait on Cement		16.00
AV - DP (fpm/mpm)				STRING WT	#REF!	New Hole Drilled		14 BOP's / Wellhead		19.50
AV - DC (fpm/mpm)				HOOK LOAD		Downhole Losses		15 Test BOP / Formation	19.00	53.25
SPP (psi/atm)				WT BELOW JARS	#REF!	Surface Losses		19 Tight hole / Fishing		9.50
SPP (Calculated)				HOURS: JAR MTR	268.5 77.5	CHEMICAL USAGE		21 Completion		
PUMP DATA				DRAG: UP DOWN		AMC Biocide G	1	21.1 Well Control		
#1: NAT 8-P-80	STROKE	9.0"		TORQUE: ON OFF		AMCPacLV	1	21.2 Other	1.00	56.25
RATE				SURVEYS: MD INC' AZ'		Soda Ash	1	TOTALS	24.00	1205.25
LINER	6.0"			2108.8 66.9 5.6		Xan-Bore	1	NON PRODUCTIVE		715.75
#2: NAT 8-P-80	STROKE	9.0"				Hydro 327	1	PRODUCTS: USED REC ON SITE		
RATE								DIESEL (L)	7200	30000 53500
LINER	6.0"							BARITE (T)		20.0
#3: IDECO T-800	STROKE	8.5"						DRILL WATER	-90000	90000 180000
RATE								DAILY MUD COSTS		\$311.62
LINER	6.0"							CUM. MUD COSTS		\$95,982.87

HOURLY OPERATIONS SUMMARY 0000 to 2400		
From	To	[IADC Code] Description
0:00	1:15	[15] PJSM. Rigged up for pressure testing standpipe and kelly cocks.
1:15	3:30	[15] Pressure tested standpipe and standpipe valves to 250psi for 5 mins and 2500psi for 10mins.
3:30	7:00	[15] Made up cup tester assembly and pressure tested annular, inner and outer kill and choke line valves to 250psi for 5 mins and 2500psi for 10ins.
7:00	9:15	[15] Reconfigured test assembly. Conducted stand alone test of annular to 250psi and 2500psi. Reconfigured for choke manifold test.
9:15	12:00	[15] Pressure tested pipe rams and all choke manifold valves to 250psi for 5 mins and 2500psi for 10mins.
12:00	15:45	[15] Re-tested blooie line 7 1/6" valve at 250psi for 15mins and 2500psi for 15mins. Re-tested annular at 250psi for 5 mins and 2500psi for 10 mins. Good tests.
15:45	16:15	[7] Serviced rig. Changed out TDS grabber box dies.
16:15	17:45	[7] Cut and slipped 80' drilling line. Re-calibrated hook load, and re-set crown and floor savers.
17:45	18:45	[21.2] Changed out spare RCD element and racked back in mast.
18:45	20:30	[15] Several attempts to pressure test kelly cocks against FOSV under side entry sub - not holding. Broke down FOSV assembly and made up assembly with combo tool.
20:30	22:00	[15] Pressure tested kelly cock and FOSV to 250psi for 5 mins and 2500psi for 10mins (2 tests). Broke out test assembly and laid out tools for offline testing.
22:00	0:00	[6.3] Made up bit and directional BHA.

HOURLY OPERATIONS SUMMARY 0000 to 0600 on 11/10/18		
0:00	3:00	Tripped in with directional BHA to 581m.
3:00	4:00	Function tested deluge system. Shut in well. Pressured up to open DDV at 730psi. Bled well down and opened 7 1/16 valve and annular.
4:00	6:00	Tripped in with directional BHA from 581m to 1177m.

POB-51: Ensign - 20, CTP - 4, OICS - 4, Halliburton - 10, OCS - 1, Site Med - 1, RMN - 1, WFT - 8, Kinetic - 2										36,300 Hrs
RECEIVED:					DISPATCHED: Liner running tool, Commander 500 cement head, 2 X 6 3/4" Jars, 2 x 6 3/4" fluid hammers, Transco 9 5/8" motor, 3 X 12 1/4" stabilisers, 1 x 17 1/2" stabiliser.					
MAXIMUM GAS:			BACKGROUND GAS:			CONNECTION GAS:			TRIP GAS:	
SUPERVISOR: Andy Phillis / Dave York			GEOLOGIST: Phil Allen / Peter Morris			RIG MGR: Gary Hender / Adam Delanie				

POB-50: Ensign - 19, CTP - 4, OICS - 4, Halliburton - 10, OCS - 1, Site Med - 1, RMN - 1, WFT - 8, Kinetic - 2				36,900 Hrs
RECEIVED:		DISPATCHED:		
MAXIMUM GAS:		BACKGROUND GAS:		CONNECTION GAS:
TRIP GAS:				
SUPERVISOR: Andy Phillis / Dave York		GEOLOGIST: Phil Allen / Peter Morris		RIG MGR: Gary Hender / Adam Delanie

			DAILY DRILLING REPORT				REPORT # 054		13/10/2018			
							SPUD: 21/08/2018		DAY # 100			
WELL	Palm Valley 13	24:00 MD/ TVD	2242 / 1989m BRT	DRILLED	11m	AFE DAYS / COST	46	\$13,467,756				
RIG	ENSIGN 932	FORMATION	Pacoota Sandstone	DAILY COST	\$106,682	CUM.	\$12,418,371					
DAILY OP'S SUMMARY		Drilled 6 1/8" hole to 2242m. (Well TD) Tripped out of hole to break down and lay out directional drilling tools. Made up scraper BHA and tripped in to 134m.										
FORECAST OP'S		Trip out with scraper. Lay down drill pipe.										
LAST CASING	178mm	7.0"	SET AT	1842.0m	FIT	12.0 EMW	MAASP	906psi	BOP TEST	10/10	DUE	24/10
LAST L.T.I. 483 Days		SAFETY						WEATHER: Day				
		1. PTSM: Drilling ahead. Well test. Review BBS cards.						Fine				
		2. PTSM: Tripping. Bleeding floats. Review BBS cards.						Night Fine				

BIT INFORMATION			BHA # 22		MUD PROPERTIES		OPERATION	HRS	CUM
WOB(Klb)	JET V(fps)		TOOL	LENGTH	Mud Type	Stored mud	1 Move/ Rig Up/Down		1122.75
RPM	H S I		Bit - Tri Cone	0.18	Depth (m)	2242	2 Drilling	5.00	510.75
BIT NUMBER	20RR16	19	Casing Scraper	1.20	Density (ppg)	9.10	3 Wash / Ream		26.25
Size (inches)	6.125	6.125	Bit Sub w/- Float	0.82	ECD (ppg)		4 Coring		
Make	Reed	Ulterra	Crossover Sub	1.08	Temp (° C)		5 Circ & Condition	3.75	66.25
Type	TK63	U71M-DEGU	25x Heavy Weight DP	235.12	Viscosity (sec)	39	6 Tripping- Bit / Casing		113.50
IADC Code					PV / YP (cp/lb)	10 / 16	6.1 Tripping- Other	10.00	114.25
Serial Number	A250214	41000			Gels (10s/m)	6 / 7	6.2 Wiper Trip		2.00
T.F.A. (in)	1.178	0.994			API Filt. (cc)	15.8	6.3 Handle BHA	2.50	86.00
Depth In (m)	2133	2022			Sand (% Vol)	tr	7 Service / Slip Line		19.75
Depth Out (m)	2242	2133			KCI (%)	3.5	8 Repairs		78.00
Total Meters	109	111			pH (strip)	9.5	10 Survey		2.75
Hours IADC OnBtm	41 35.1	52.8 44.3			LGS %/Vol	4.3	11 Logging		11.50
ROP IADC OnBtm	2.7 3.1	2.1 2.5			Chlorides (ppm)	18000	12 RU for Csg / Cmt		30.00
Condition Out	3 3 BT N X 0 WT TD				Surface Vol. (Bbls)	1120	12.1 Run Casing		42.50
FLOW DATA			BHA LENGTH	238.40	Hole Vol.	371	12.2 Cementing		15.25
RATE (gpm/lps)			BHA WEIGHT	18.85	Mud Mixed	374	13 Wait on Cement		16.00
AV - DP (fpm/mpm)			STRING WT	108.35	New Hole Drilled	1.3	14 BOP's / Wellhead		20.75
AV - DC (fpm/mpm)			HOOK LOAD	35.00	Downhole Losses		15 Test BOP / Formation		53.25
SPP (psi/atm)			WT BELOW JARS		Surface Losses	304	19 Tight hole / Fishing		9.50
SPP (Calculated)			HOURS: JAR MTR	309.5 77.5	CHEMICAL USAGE		21 Completion		
PUMP DATA			DRAG: UP DOWN		AMCPacLV	1	21.1 Well Control		
#1: NAT 8-P-80	STROKE	9.0"	TORQUE: ON OFF		Soda Ash	3	21.2 Other	2.75	59.00
RATE			SURVEYS: MD INC' AZ'		Xan-Bore	3	TOTALS	24.00	1277.25
LINER	6.0"		2217.4 80.8 4.8		Hydro 327	7	NON PRODUCTIVE		716.75
#2: NAT 8-P-80	STROKE	9.0"	2242 82 4.5		AMC Superfoam	36	PRODUCTS: USED REC ON SITE		
RATE							DIESEL (L)	9850	27350
LINER	6.0"						BARITE (T)		20.0
#3: IDECO T-800	STROKE	8.5"					DRILL WATER	100000	30000
RATE							DAILY MUD COSTS \$3,001.33		
LINER	6.0"						CUM. MUD COSTS \$106,904.99		

HOURLY OPERATIONS SUMMARY 0000 to 2400			
From	To	[IADC Code]	Description
0:00	3:15	[2]	Directionally drilled 6 1/8" hole, Rotating, from 2231m to 2237m MD. Air: 2000scfm at 1100psi, Mist: 20gpm, WOB: 10-18 klbs. WHP 15psi Av ROP: 1.8 m/hr
3:15	6:00	[5]	Air pack booster dropped out. Annulus became heavily loaded up with foam. Work pipe while unloading well and repair booster.
6:00	7:15	[21.2]	Changed out RCD bearing. (leaking)
7:15	9:00	[2]	Directionally drilled 6 1/8" hole, Rotating, from 2237m to 2242m MD. Air: 2000scfm at 1100psi, Mist: 20gpm, WOB: 10-18 klbs. WHP 15psi Av ROP: 4m/hr
9:00	10:00	[5]	Blew hole clean at 2242m.
10:00	11:15	[6.1]	Tripped out from 2242m to 1784m.
11:15	12:15	[6.1]	Made up pressure release tool and bled down string.
12:15	14:45	[6.1]	Tripped out from 1784m to 580m.
14:45	17:45	[6.1]	Shut in well. Close DDV. Inflow indicates DDV bypassing. Flushed DDV. DDV holding. DDV flushed at 760psi.
17:45	18:45	[6.3]	Rack back RCD stand. Laid out jars. Changed elevator inserts.
18:45	20:00	[6.1]	Tripped out from 580m to 24m.
20:00	21:30	[6.3]	Broke out bit and directional tools and laid out.
21:30	23:00	[21.2]	Changed out RCD rubbers.
23:00	0:00	[6.1]	Made up scraper BHA and tripped in to 134m.

HOURLY OPERATIONS SUMMARY 0000 to 0600 on 14/10/18			
From	To	[IADC Code]	Description
0:00	1:45		Tripped in with casing scraper to 555m.
1:45	2:30		Shut in to pressure up well to open DDV at 737psi. Bled well down and opened well.
2:30	5:30		Tripped in with casing scraper from 555m to 1775m.
5:30	6:00		Break circulation and work scraper.

POB-49: Ensign - 20, CTP - 4, OICS - 4, Halliburton - 8, Site Med - 1, RMN - 1, WFT - 8, Kinetic - 3								38,064 Hrs	
RECEIVED: 1 X set tubing slip elevators.						DISPATCHED:			

MAXIMUM GAS:		BACKGROUND GAS:		CONNECTION GAS:		TRIP GAS:	
SUPERVISOR: Andy Phillis / Dave York		GEOLOGIST: Phil Allen / Peter Morris		RIG MGR: Simon Luke Bryant / Adam Delanie			

POB-52: Ensign - 20, CTP - 3, OICS - 4, Halliburton - 8, Site Med - 1, RMN - 1, WFT - 8, Kinetic - 3, IOT - 4				38,688 Hrs
RECEIVED:			DISPATCHED:	
MAXIMUM GAS:		BACKGROUND GAS:		CONNECTION GAS:
SUPERVISOR: Andy Phillis / Dave York		GEOLOGIST: Phil Allen / Peter Morris		RIG MGR: Simon Luke Bryant / Adam Delanier

POB-49: Ensign - 20, CTP - 4, OICS - 4, Halliburton - 4, Site Med - 1, RMN - 1, WFT - 8, Kinetic - 3, IOT - 4				39,276 Hrs
RECEIVED:		DISPATCHED:		
MAXIMUM GAS:		BACKGROUND GAS:		CONNECTION GAS:
TRIP GAS:				
SUPERVISOR: Andy Phillis / Dave York		GEOLOGIST: Phil Allen / Peter Morris		RIG MGR: Simon Luke Bryant / Adam Delanie

			DAILY DRILLING REPORT			REPORT # 058		17/10/2018				
						SPUD: 21/08/2018		DAY # 104				
WELL	Palm Valley 13	24:00 MD/ TVD	2242 / 1989m BRT	DRILLED		AFE DAYS / COST	46	\$13,467,756				
RIG	ENSIGN 932	FORMATION	Pacoota Sandstone		DAILY COST	\$113,633	CUM.	\$13,022,529				
DAILY OP'S SUMMARY		Ran 3 1/2" Tubing & Landed same .Completed seal tests as per programme.Ran Kinetic wireline & Displaced tubing to air as per programme .Nippled down BOP.Nippled up Xmas Tree & pressure tested as per programme										
FORECAST OP'S		Release Rig & Rig down to move										
LAST CASING	178mm	7.0"	SET AT	1842.0m	FIT	12.0 EMW	MAASP		BOP TEST	10/10	DUE	24/10
LAST L.T.I. 487 Days		SAFETY		1. PTSM: Review BBS cards. Run Tubing. Pressure testing. Slick line operations. 2. PTSM: Review BBS cards. Slick line operations. Nipple down BOP.				WEATHER: Day		Fine		
								Night		Fine		

BIT INFORMATION					MUD PROPERTIES		OPERATION		HRS	CUM
WOB(Klb)	JET V(fps)		TOOL	LENGTH	Mud Type					
RPM	H S I				Depth (m)					
BIT NUMBER	20RR16	19			Density (ppg)					
Size (inches)	6.125	6.125			ECD (ppg)					
Make	Reed	Ulterra			Temp (° C)					
Type	TK63	U71M-DEGU			Viscosity (sec)					
IADC Code					PV / YP (cp/lb)					
Serial Number	A250214	41000			Gels (10s/m)					
T.F.A. (in)	1.178	0.994			API Filt. (cc)					
Depth In (m)	2133	2022			Sand (% Vol)					
Depth Out (m)	2242	2133			KCl (%)					
Total Meters	109	111			pH (strip)					
Hours IADC OnBtm	41 35.1	52.8 44.3			LGS %/Vol					
ROP IADC OnBtm	2.7 3.1	2.1 2.5			Chlorides (ppm)					
Condition Out	3 3 BT N X 0 WT TD				Surface Vol. (Bbls)					
FLOW DATA			BHA LENGTH		Hole Vol.	416				
RATE (gpm/lps)			BHA WEIGHT		Mud Mixed					
AV - DP (fpm/MPM)			STRING WT		New Hole Drilled					
AV - DC (fpm/MPM)			HOOK LOAD		Downhole Losses					
SPP (psi/atm)			WT BELOW JARS		Surface Losses	270				
SPP (Calculated)			HOURS: JAR MTR	309.5	77.5	CHEMICAL USAGE				
PUMP DATA			DRAG: UP DOWN							
#1: NAT 8-P-80	STROKE	9.0"	TORQUE: ON OFF							
RATE			SURVEYS: MD INC' AZ'							
LINER	6.0"		2242 82 4.5							
#2: NAT 8-P-80	STROKE	9.0"								
RATE										
LINER	6.0"									
#3: IDECO T-800	STROKE	8.5"								
RATE										
LINER	6.0"									
HOURLY OPERATIONS SUMMARY 0000 to 2400										
From	To	[IADC Code] Description								
0:00	0:30	[21] PJSM for running 3 1/2" tubing. Ran total of 193jts 3 1/2" EUE 9.3lb/ft L80 tubing to tag packer at 1797.37mm								
0:30	1:00	[21] Tagged packer with 3klbs. Tool latched. Unlatched tool and confirmed space out. Laid out one joint of tubing. String weight 46klbs up, 45klbs down.								
1:00	2:00	[21] Made up landing joint with tubing hanger and landed tubing into packer with 20klbs squat and 0.5m travel after initial tag up. Confirmed latch with 10klbs overpull								
2:00	4:30	[21] Confirmed hanger landed via "View Port". Made up all tie down bolts. Pressure tested tubing seals at 5000psi for 15mins for each of two tests.								
4:30	7:30	[21] Backed out and pulled landing joint. Closed blind rams and pressure tested tubing to 2000psi for 15mins. Pressure tested annulus to 2000psi for 10mins.								
7:30	10:00	[21] Rigged up slick line equipment and pressure tested lubricator to 2000psi for 10 mins.								
10:00	12:00	[21] Ran in slick line and opened SSD. Pulled slick line into lubricator. Displaced tubing with air for 51bbls returns.								
12:00	13:00	[21] Ran in slick line and closed SSD. Pulled slickline and bled down tubing pressure.								
13:00	15:00	[21] Rigged down slick line. Pulled out 3 1/2" landing joint and installed 3" TWC.V.Pressure test Annulus to 1000psi								
15:00	0:00	[21] Removed blooie line. Nippled down BOP. Cleaned Mud Tanks								
HOURLY OPERATIONS SUMMARY 0000 to 0600 on 18/10/18										
0:00	2:00	Continue Nipple Down BOP								
2:00	6:00	Install & Nipple up Xmas Tree .Pressure test connection to 3000psi for 15 mins.Test neck seals/Void 3000psi for 10 mins .Test TWC.V/Tree 3000psi for 10 mins								
POB-46: Ensign - 20, CTP - 2, OICS - 4, Halliburton - 5, Site Med - 1, WFT - 7, Kinetic - 3, IOT - 4										
RECEIVED:										40,416 Hrs
DISPATCHED:										
MAXIMUM GAS:			BACKGROUND GAS:			CONNECTION GAS:			TRIP GAS:	
SUPERVISOR: Andy Phillis / Dave York			GEOLOGIST:			RIG MGR: Simon Luke Bryant / Adam Delanie				

